

How Spillovers Erode Participation in Conservation Buybacks: Evidence from the Texas Inshore Shrimp Fishery

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Abstract

Buyback programs are widely used to reduce overcapitalization and generate social benefits in settings such as land retirement, groundwater conservation, and fisheries. Yet evaluations often stop at program boundaries, even though removals can generate spillovers: displaced activity may move elsewhere, while those who remain benefit from a replenished resource. I study how these gains feed back into buyback program participation. As capacity is removed, the continuation value of remaining participants rises, creating an incentive to hold out rather than exit. Successful removals can therefore weaken the program participation required for further success.

I quantify this feedback-loop using the Texas Parks and Wildlife Department's 26-year inshore shrimp license buyback as a testbed. I estimate a two-stage structural model of vessels' spatial fishing choices and license-retirement decisions, then use it to simulate continuation values and reconstruct harvest and auction activity when vessel-level data are unavailable. I find that for every six active license-holders who bid, roughly one who would otherwise have participated holds out as stock recovery makes staying more valuable. Program effectiveness is further constrained by two margins: approximately three-quarters of retired licenses were already inactive, and surviving vessels replace more of the retired harvest in denser bay systems. Restricting purchases to active licenses would have retired 80% more trawl-days per license (307 against 171). Together, the results show that conservation buybacks can undermine participation and additionality through the benefits they create for those who remain.

Contents

1	Introduction	1
2	Institutional Background	5
2.1	The Texas Inshore Shrimp Fishery	5
2.2	The Texas Shrimp License Buyback Program	8
3	Data	12
4	A Dynamic Model of Common-Pool Exploitation	16
4.1	Model Overview	16
4.2	Production Function and Stock Dynamics	19
4.2.1	Harvest Production	19
4.2.2	Latent Stock Index	20
4.2.3	Within-Season Growth Function	21
4.3	Bay System Choice	21
4.4	State Variable Transitions	23
4.5	Equilibrium and Continuation Values	24
4.5.1	Markov Perfect Equilibrium	24
4.5.2	Forward Simulation	25
4.5.3	Continuation Value Computation	26
5	Estimation Results: Fishery Dynamics	28
5.1	Production Function Estimates	28
5.2	Stock Dynamics	29
5.3	State Transition Fit	32
5.4	Stage 1 Conditional Choice Probabilities	32
5.5	Structural Profit Parameters	32
5.6	Continuation Values and Auction Entry	36
6	Counterfactual Analysis: Spillovers, Offsets, and Targeting	38
6.1	Overview	38
6.2	The Synthetic Fleet and Full-Horizon Back-Cast	39
6.3	The Participation Distortion and its Decomposition	42
6.4	The Structural Offset	45
6.5	The Cost-Effectiveness of Removal Rules	48
6.6	Robustness	50
7	Conclusion	52
A	Supplementary Estimation Results	61
B	Synthetic-Fleet Construction and Full-Horizon Back-Cast	66
C	Inference for the Counterfactual	69

D Counterfactual Supplementary Exhibits	70
E Model Validation	72

1 Introduction

Buyback programs—reverse auctions and fixed-price offers that pay resource users to retire environmentally harmful capital—have become a mainstream conservation instrument. Programs in the United States alone have spent billions each year on cropland retirements [U.S. Department of Agriculture, Farm Service Agency, 2025a], and millions on several fishing license buyback programs [NOAA Fisheries, 2024a] and groundwater rights retirements [Sakas, 2022]. Their appeal over other instruments such as taxes and quotas is that they are voluntary: participants reveal, and are compensated for, their privately known cost of exit, relieving an overcapitalized or economically fragile resource without forcing out vulnerable users [Ferraro and Kiss, 2002, Jack et al., 2008]. But when participation falls short, the social benefits fall with it, delivering far less per dollar when participant pools thin [DePiper, 2015, Rolfe et al., 2018]. When the Conservation Reserve Program (CRP) falls 1 million acres short of its statutory cap [U.S. Department of Agriculture, Farm Service Agency, 2025a], or recruits’ retirements that are only 25% additional [Aspelund and Russo, 2026], it may be read as a failure of price, outreach, or targeting. But what if withholding participation is a rational response to the program’s own success?

A large literature accounts for how the effects of conservation programs spill beyond their targets, often called “slippage” or “leakage” effects. For example, when a nation commits to a carbon abatement goal, non-committed countries gain a competitive advantage in carbon-intensive industries [Aichele and Felbermayr, 2015, King and van den Bergh, 2021]; and when a landowner sells cropland for non-farm uses, they are likely to bring other idle land back into farming [Wu, 2000]. Pfaff and Robalino [2017] organize these mechanisms into several channels, including input reallocation, market prices, learning, and non-pecuniary, which each elicit a behavioral response. A fifth channel, “ecological-physical links”, is largely set aside because it “... does not involve human behavior”, though the authors note that “... those links may also lead to further behavioral response and thus additional program spillovers.” Examples include ecosystem-service gains to neighboring farms when cropland is

retired to grassland [Landis et al., 2008], increasing pumping by connected groundwater users after nearby rights are retired [Rouhi Rad et al., 2021, Ekpe and Klis, 2023], and reallocated fishing activity after quota programs restrict harvest in a primary fishery [Cunningham et al., 2016] (See Table 1 for more examples). This paper focuses on this fifth channel, but on a margin that has received little attention: ecological recovery changes the private continuation value of agents who remain eligible for the program. I ask how that endogenous gain feeds back into participation, and how program design can account for it.

While the programs above generate positive externalities for those who remain, auction theory, which serves as the conceptual basis for many analyses in this space, has more often studied strategic distortions arising from negative externalities. When a rival’s win is costly, bidders compete more aggressively, which can increase the principal’s revenue [Hu et al., 2013]. This logic appears in settings such as timber rights [Kuehn, 2019], research patents [Kwon, 2020], and security auctions [Hernandez-Chanto and Fioriti, 2019], where winning confers an advantage in downstream competition. In a conservation buyback, the sign of the externality reverses: when a rival’s exit improves conditions for those who remain, each participant would prefer that someone else sell instead. Positive externalities can therefore weaken participation and raise the cost¹ of inducing further exit [Calel, 2012, Espinola-Arredondo, 2008, Choi et al., 2018]. The same holdout logic appears in models of voluntary environmental agreements, where the incentive to remain outside a coalition grows as more countries commit to abatement [Barrett, 1994, Nordhaus, 2015].

Because voluntary procurement compensates agents for their private cost of exit, and that cost is lowest for those who would have scaled back regardless, a parallel concern emerges when a program’s budget pays for conservation that would have happened without it. This adverse selection is the central efficiency problem in conservation contracting: it has been studied as contract design under asymmetric information [Ferraro, 2008], screening

¹In a forward auction, bids reflect willingness to pay and the principal seeks high bids; in a reverse auction, asks reflect willingness to sell and the principal seeks low asks. The strategic implications of externalities therefore translate with the direction of the mechanism.

when additionality is unobserved [Mason and Plantinga, 2013], and measurement against a counterfactual of rejected bids [Rosenberg and Pratt, 2024, Aspelund and Russo, 2026]. This work treats additionality as a property of the individual contract, but when benefits spillover on agents who remain the addition of a retirement becomes endogenous. But in a common-pool resource, the conservation delivered is attenuated further by how much of the resource the remaining users recapture, so even a removal that is fully additional in the usual sense may retire less than it appears.

This recapture, the offset, is itself a familiar object. It has been measured as reduced-form leakage in fisheries [Cunningham et al., 2016] and modeled as the redistribution of effort across patches in spatial bioeconomics [Sanchirico and Wilen, 2020], and the buyback policy literature has long warned that these programs are blunted by the anticipation of future removals [Clark et al., 2005], by the replacement of retired capacity [Holland et al., 1999], or by fishermen overinvesting in gear in anticipation of a thinned rival pool, known as “capital stuffing” [Clark and Munro, 2002]. The same recapture exists wherever the rights are spatial and the resource is shared, from effort displacement in rights-based fisheries [Abbott and Wilen, 2009] to pumping externalities across a common aquifer [Pfeiffer and Lin, 2012].

Several issues make identification in this vein challenging. For example, the debate on the magnitude of physical spillovers in the CRP has persisted for decades, with credible estimates ranging from null to substantial depending on how econometricians choose to handle issues like land heterogeneity, endogenous entry, and unobservable variables [Wu, 2000, Roberts and Bucholtz, 2005, Fleming, 2014]. One particular challenge emerges from the wide breadth of spillover channels outlined by Pfaff and Robalino [2017]: it is challenging to isolate congestion, price, and physical links when entry into programs is endogenously determined by an individual’s continuation value. To bridge this empirical gap, I frame the Texas Parks and Wildlife Department’s (TPWD) inshore shrimp license buyback, a reverse auction run over 41 rounds and 26 years across eight bay systems, as a testbed. This setting provides two unique comparative advantages for identification over previous buyback

settings. First, the spillover effect is tractable and structurally identifiable; the removal of licenses leads to a decrease in cumulative harvest, which in turn improves stock health and allows remaining vessels to harvest more shrimp at lower cost. Second, it isolates a biological spillover. Congestion relief could in principle generate its own positive spillover, but the bay systems span thousands of square kilometers and host numerous other fisheries; I show using bay-specific shift-share instruments that the congestion effect is negligible and insignificant. Price-mediated leakage is likewise minimal, as Gulf-harvested shrimp make up only 4.5% of the U.S. food-shrimp market [Liese et al., 2026].

First, I provide evidence that a positive stock externality can distort participation within a repeated conservation buyback. Leakage and slippage are familiar in conservation policy: program gains erode as activity shifts to unregulated places, agents, or periods [Wu, 2000, Cunningham et al., 2016]. The literature primarily studies spillovers outward from the program boundary. I instead study a feedback loop within the program itself: by removing harvest, the buyback rebuilds the stock and raises the continuation value of vessels that remain, making future exit less attractive. The TPWD setting allows me to isolate this biological channel from price and congestion effects. A one-million-pound increase in cumulative seasonal harvest lowers stock health by 18%; a one-util increase in the continuation value of a license lowers the probability of bidding by 7%; and over the full program, stock recovery suppresses participation by about 13.9%.

Second, I extend the dynamic discrete-choice framework for common-pool exploitation of Aguirregabiria and Mira [2002, 2007], brought to fisheries by Huang and Smith [2014], to a spatial, multi-bay setting, and use the estimated model to reconstruct participation and harvest over the program’s full twenty-six-year horizon, including years for which vessel-level activity data is unavailable. Third, I unify additionality and offset into a single measure of conservation delivered per dollar, and derive the offset margin structurally from the resource externality itself. Asset retirement falls short on two margins. The first is extensive and familiar from land-retirement programs: a purchased license may have been inactive to

begin with, so its retirement is non-additional [Ferraro, 2008, Jack et al., 2008]. The second is intensive and specific to a common-pool resource: even a fully active retirement delivers less than one-for-one conservation because surviving vessels fish the freed stock and recapture part of the removed harvest. I recover this offset directly from the estimated depletion technology and show that it is not constant, but rises with local fleet density from 0.04 in the thinnest bay to 0.26 in the densest as recapture saturates with remaining activity. Together, these margins imply large heterogeneity in effective additionality and conservation per dollar: screening purchases for recent activity and targeting bays by offset and seller activity can substantially increase net trawl-days retired for the same budget.

The paper proceeds as follows. Section 2 describes the Texas inshore shrimp fishery and the buyback program. Section 3 details the data. Section 4 develops the two-stage structural model with harvest production, latent stock dynamics, state-variable transitions, and the equilibrium used to simulate continuation values. Section 5 reports estimates and traces the spillover-to-participation chain. Section 6 simulates counterfactual program designs and isolates the participation distortion from retirement-driven stock replenishment; Section 7 concludes, and supporting tables and figures are listed in Section A.

2 Institutional Background

2.1 The Texas Inshore Shrimp Fishery

The Texas inshore shrimp fishery, regulated by the Texas Parks and Wildlife Department (TPWD), is composed of eight major bay systems along the Gulf of Mexico, comprising an area of roughly 4,363 square kilometers (see Figure 1). Roughly 293 million pounds of bay shrimp tails have been landed since 1981, generating a cumulative total revenue of \$1.2 billion (in adjusted 2023 US dollars). While these bay systems are large, operations are still heavily interconnected between vessels. Shrimp populations need specific levels of salinity and narrow corridors to thrive. Biomass stock can also be heavily influenced by

Table 1: Buyback Settings, Spillovers, and Feedback Loops

Setting (retired unit)	Spillover	Recapture and participation feedback	Flagship voluntary program (approx. scale)
Marine & coastal fisheries (licenses, vessels)	Removal slows depletion, raising the stock. <i>This paper</i> : depletion semi-elasticity ≈ 0.20	Remaining vessels intensify, rising with density (0.03–0.29).	NOAA capacity buybacks: Pacific groundfish $\approx \$46\text{M}$, AFA pollock $\approx \$75\text{M}$ [NOAA Fisheries, 2024b]; TX inshore shrimp $\approx \$15\text{M}$ (<i>this paper</i>)
Groundwater (acreage, pumping rights)	Less pumping raises the water table, cutting cost and raising yield for connected users	Neighbors pump more [Rouhi Rad et al., 2021].	CREP (Republican River); SGMA following $\approx 0.5\text{--}0.9\text{M}$ ac [Public Policy Institute of California, 2023]; CO buy-outs $\approx \$60\text{M}$ [Sakas, 2022]
Surface water / river flow (entitlements, diversions)	Retired diversions leave more in-stream and downstream flow	Remaining and junior users take the freed water.	Murray–Darling Basin water buybacks $\approx \text{A\$}2.5\text{B}$, $\sim 1,200$ GL [Australian Government Department of Climate Change, Energy, the Environment and Water, 2024]
Rangeland / grazing (permits, AUMs)	Less grazing improves forage and riparian condition for remaining permittees	Remaining herds expand.	Voluntary Grazing Permit Retirement (US; mostly proposed) [U.S. House of Representatives, 2025]
Cropland: pollination & pest control (cropland to habitat)	Habitat raises pollination and natural pest control on adjacent farms (local, diffuse)	Slippage onto the owner’s other land [Wu, 2000].	CRP, incl. pollinator practice CP42, $\approx \$1.6\text{--}1.8\text{B/yr}$, $\sim 23\text{M}$ ac [Aspelund and Russo, 2026]
Cropland / watershed: water quality (cropland, buffers)	Less runoff yields cleaner water and less sediment downstream	Slippage.	CREP riparian buffers; Chesapeake Bay nutrient trading [U.S. Department of Agriculture, Farm Service Agency, 2025b]
Wetlands (cropland to restored wetland)	Flood attenuation, water quality, and habitat for neighbors	Limited.	Wetland Reserve / ACEP-WRE, $\sim 2\text{--}3\text{M}$ acres [U.S. Department of Agriculture, Natural Resources Conservation Service, 2025]
Carbon / emissions (emissions, allowances, credits)	Avoided emissions, a non-rival climate benefit enjoyed regardless of participation	Geographic leakage [Aichele and Felbermayr, 2015, King and van den Bergh, 2021]. Feedback: absent (competitiveness channel separate)	EU ETS; RGGI; California cap-and-trade; CDM / REDD+ and voluntary off-sets

Notes: Acronyms, in alphabetical order. ACEP-WRE: Agricultural Conservation Easement Program, Wetland Reserve Easements (USDA). AFA: American Fisheries Act. AUM: animal unit month, the forage required by one cow-calf pair for one month. CDM: Clean Development Mechanism. CP42: CRP conservation practice 42, pollinator habitat. CREP: Conservation Reserve Enhancement Program. CRP: Conservation Reserve Program. EU ETS: European Union Emissions Trading System. GL: gigaliter, one billion liters. NOAA: National Oceanic and Atmospheric Administration. REDD+: Reducing Emissions from Deforestation and Forest Degradation, including conservation, sustainable forest management, and enhancement of forest carbon stocks. RGGI: Regional Greenhouse Gas Initiative. SGMA: Sustainable Groundwater Management Act (California). A\$ denotes Australian dollars; CO and TX denote Colorado and Texas.

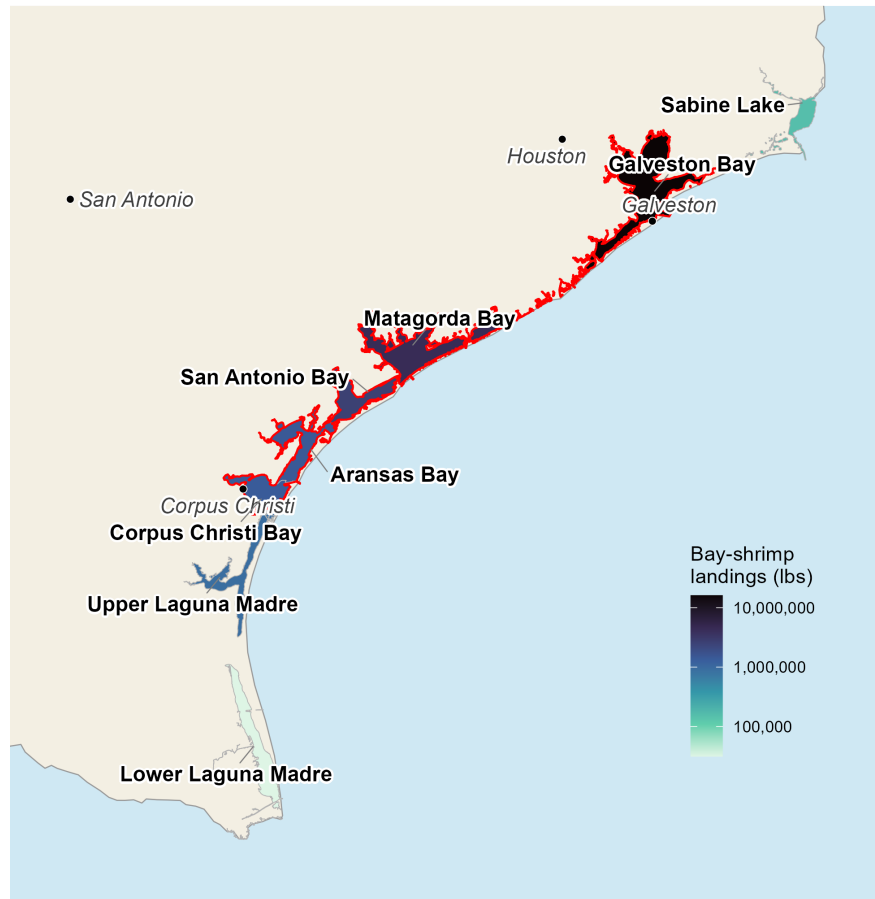


Figure 1: Texas Inshore Shrimp Bay Systems. Bay-shrimp landings (lbs) are displayed from 2008 to 2023 ticketed Bay shrimp landings. Five bay systems outlined in red are the focal point for this paper: Galveston, Matagorda, San Antonio, Aransas, and Corpus Christi bay systems (read North to South).

active licenses, as less-exploited populations can thrive. While vessels are not bumping into each other, they generally can be assumed to enjoy positive returns from less capitalized bay systems.

Inshore shrimp refers to two broad species categories: “Bay” shrimp, the focus of this paper, is commercially produced for food, while “bait” shrimp is primarily used as an input toward other fisheries. Because many state laws prohibit the use of imported shrimp as bait, Gulf bait shrimp carries huge market power domestically [Texas Parks and Wildlife Department, 2026]. The bay shrimp market, on the other hand, faces substantial pressure from foreign imports such that vessels can safely be assumed to be price takers² [Liese et al., 2026]. TPWD also manages deep-sea shrimp fisheries which they refer to as “Gulf” shrimp, but they have not sought to reduce capitalization in those fisheries. The goal of this paper is to analyze bio-economic spillovers, so I focus exclusively on Bay licenses to avoid concerns of price endogeneity.

Roughly 88% of bay shrimp landings, and 84% of total trips, happen within one of two designated seasons every year: a “Spring” open season from May 15th to July 15th, and a “Fall” open season from August 15th to November 30th (see Figure 2). To avoid anomalous harvest activity on off-season days, and as an attempt to isolate commercial activity from recreational, the analysis involved in this paper focuses on in-season activity only.

2.2 The Texas Shrimp License Buyback Program

Texas inshore shrimping expanded over the latter half of the twentieth century to the point of severe over-capitalization. TPWD characterized the bay shrimp fishery as simultaneously over-fished, fully exploited, and supporting more vessels than the resource could sustainably bear [McKinney, 2004]. The agency’s entry-side response came in 1995, when the Texas Legislature established a limited-entry program for the commercial bay and bait shrimp

²Inverse demand, instrumented by weather, shows a negligibly small though statistically significant market demand elasticity of -11 , implying a 10% shock to supply leads to a 0.9% price change. The residual demand facing an individual vessel is more elastic still: see Table A.3

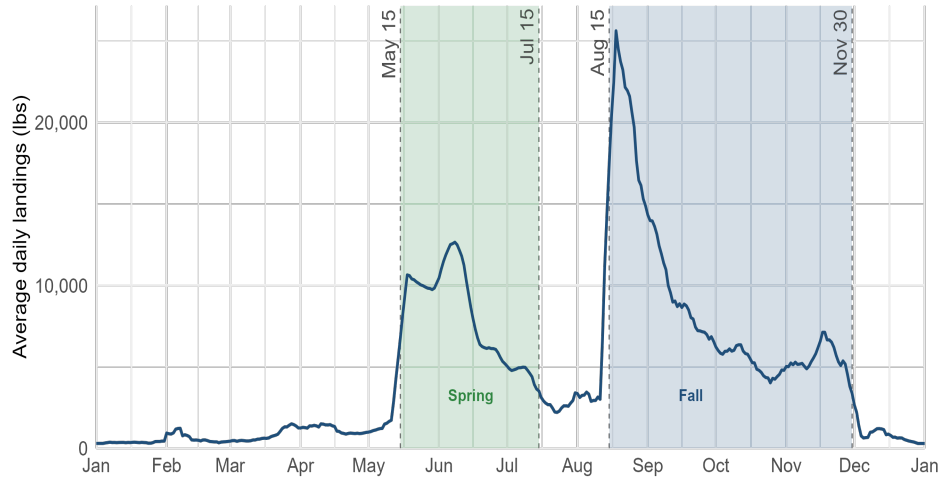


Figure 2: Daily Landings Averaged Across 2008 to 2023 (7-day rolling mean). Spring and Fall open-seasons illustrated above, from May 15th to July 15th and August 15th to November 30th, respectively.

fisheries, placing a moratorium on the issuance of new licenses and fixing the pool of active shrimp fishers.

The moratorium halted further growth in capacity but could not draw down the fleet already in place, and a secondary market of license sales between fishers prevented the desired decrease in capacity. This motivated the inclusion of additional levers to retire licenses from the now-frozen pool. Policymakers noticed, however, that the overcapitalized fishery is also quite economically fragile. National Marine Fisheries Service economic data show that although U.S. shrimp consumption has more than quadrupled between 1984 and 2023, that demand was met almost entirely by cheaper farmed imports: the Gulf fishery’s share of the domestic market fell to roughly 4.5% by 2023, and the inflation-adjusted ex-vessel price of Gulf shrimp declined from over \$6 per pound in 1984 to under \$2, its lowest level on record [Liese et al., 2026]. Additional financial pressures include volatile diesel costs, an aging fleet, and recurrent threats from tropical storms, leaving the industry economically fragile. For this reason, TPWD deliberately considered voluntary instruments alternatives that would force vessels out of production or restrict their economic capabilities. As Coastal Fisheries Division director Larry McKinney explained, the buyback was chosen as the “...

least-disruptive management tool available,” precisely because the goal was to stabilize the fishery “... socially, economically and biologically,” without further damaging an already precarious market [McKinney, 2004].

The buyback program was implemented in 1997 as a reverse auction in which bay and bait shrimp license-holders bid an asking price to voluntarily forfeit a license. At least one round (at most two) of the buyback has been run annually for every year except for 2023, with the number of purchased licenses limited by a fixed-round budget. Auctions commenced after the implementation of the 1995 moratorium on new license issuance, ensuring a fixed pool of potential entrants into perpetuity or until the policy is changed. All licensed shrimpers are eligible to participate regardless of recent harvesting activity. Of the 504 distinct license-holders who submitted bids between 2008 and 2023, 218 recorded no shrimp landings anywhere in the ticket data over that entire sixteen-year period. Inactivity is even more pronounced at the bid level: because anyone may bid repeatedly, approximately 66% of all bids during this period sought payment for licenses associated with no contemporaneous fishing activity, and therefore requested non-additional retirements. Among the 226 bay licenses TPWD actually purchases in 2008 to 2023, 24% belong to vessels that landed shrimp in the round year or the season before.

In the 26 years since its inception, the program has run over 41 rounds, retiring 2,120 of a pre-moratorium pool of 3,231 bay and bait shrimp licenses (65.56% of total licenses at the onset), at a total cost exceeding \$15 million. Effort reduction has been successful: Texas bays recorded roughly 10,000 trawl-days recorded, about half the 20,000 recorded two decades prior [Texas Parks and Wildlife Department, 2007]. Recorded landings since 2006 have fluctuated between 0.5 and 3.5 million lb tails, compared to between roughly 9 and 15 million before the buyback (see Figure 3). 92% of vessels held both Bait and Bay licenses at the beginning of the program. Vessels are allowed to hold multiple of the same license (e.g. two bay licenses), in theory to increase the limit to which they can fish, though only 0.08% of in-season trips went over the per-license daily limit of 600 from 2008 to 2023,

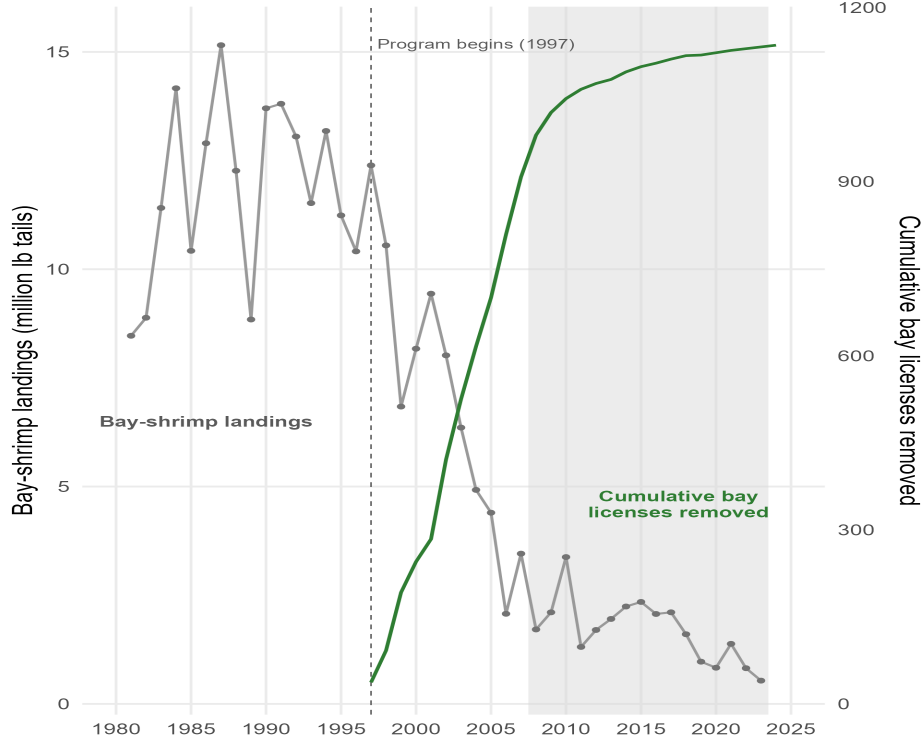


Figure 3: Texas Bay-Shrimp Landings (million lbs tails).

suggesting vessels carry this license as a transferable asset in the secondary market rather than to augment operational fishing capacity.

The auction operates as follows. In late January, license-holders choose whether or not to participate in the auction. In the auction season (typically end of peak season in mid July or early November), entered license-holders can choose to bid an asking price (\$/license) at which they would be willing to permanently forfeit their fishing rights. Participants are not informed of who they are bidding against. TPWD receives these bids and calculates a license-specific estimated market value (EMV), based on the age of the license and the length of the associated vessel, which serves as a reserve price. Bidders are aware that their EMV is conditioned on these variables, but don't know the specific formula.³ Bids are sorted by their gap (EMV - Bid), and TPWD accepts bids beginning with the largest gaps first until

³The formula used to calculate EMV was adjusted at three points, at rounds 3, 6, and 34. Each new formula increased the EMV for every license permissible in the auction to elicit more participation. This appears to have had a weak effect on increasing bid amounts: see Panel (A) of Figure 4.

either their budget for that round, or the set of positive gaps, is exhausted.⁴

License-holders whose bids are accepted may subsequently rescind their offers. This occurred for 376 of 2,619 accepted bids, or about 14%. Rescission carries no penalty for future participation, and returning bidders are treated no differently from first-time entrants. This option may reflect changes in willingness to sell between auction entry and final acceptance, but it may also facilitate strategic learning about the acceptance rule. Consistent with the latter interpretation, 38 of the 42 vessels that had a bid accepted, rescinded, and then returned in the following round and submitted a bid at least as high as before. By contrast, among vessels whose bids were rejected and who returned in a later round, a majority subsequently lowered their asks (see Panel (B) of Figure 4).

TPWD has run at least one round of the auction every year since 1997, except in 2023. Rounds were typically held at end of season, though many years held an additional round at the end of the spring season.⁵ The bidder pools started large, fluctuating between roughly 75 and 150 bidders, until an eventual decline after round 18 (see Panel (C) of Figure 4). The types of entrants has also changed, with the average vessel length associated with a submitted bid increasing from 35 feet in round 1 to 45 feet by round 40 (see Panel (D) of Figure 4). This compositional change may reflect a thinning of the marginal vessels in operation, capitalization on a thinner rival vessel pool (also known as “capital stuffing”), or general industry trends.

3 Data

TPWD provided vessel-level trip-ticket records from 2008 to 2023, reporting each commercial landing by vessel identifier, date, species (bay or bait shrimp), bay system, landed weight of caught shrimp, and shrimp prices by bay system. I refer to these records throughout as the ticket data. While 8 bay systems exist in the dataset, the majority of shrimp landings

⁴Rejected license-holders are not informed whether their bid was declined due to an insufficient gap relative to rivals or because the round’s budget was exhausted before their bid was reached.

⁵Years with two rounds held included 1997-99, 2002-05, 2007-11, and 2015.

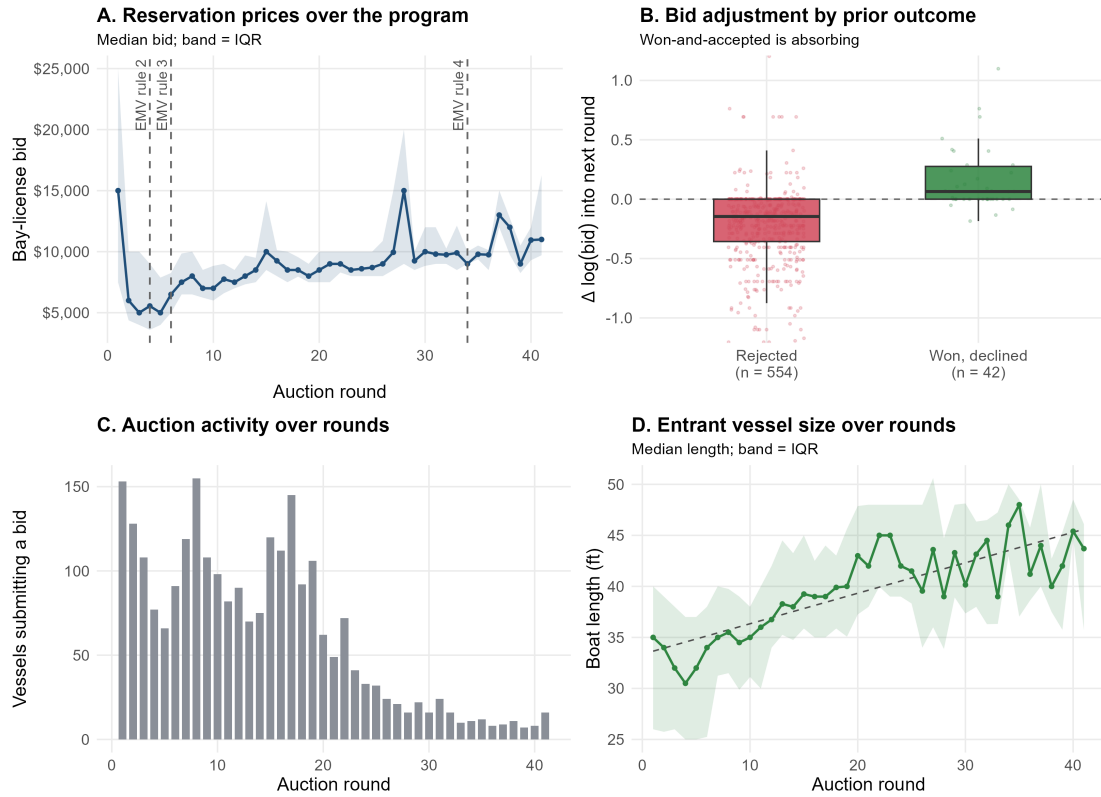


Figure 4: Summary Figures on the Buyback Auction

Note: Four figures above summarize auction activity. Panel (A) shows placed bids per round, with vertical lines denoting when TPWD changed the estimated market value of a license used as the reserve price, i.e. the maximum accepted price of a license based on vessel length and license age. Panel (B) shows difference in bid amount conditional on prior-round bid. Panel (C) shows bidder pools over time. Panel (D) shows boat length over time.

occur in 5 bay systems (92%). I focus exclusively on the Galveston, Matagorda, Aransas, Corpus Christi, and San Antonio Bay systems (see Figure 1). The trip tickets serve two roles, as both a bay-day harvest series for the stock and production models and vessel-level daily location choices, while the auction records supply round-by-round participation and bid outcomes. Unfortunately, trip-ticket data at the vessel level only exists in years 2008 to 2023 (rounds 20 to 40), which excludes some of the years with the largest bidder pools.

I use the TPWD boat/motor ownership inquiry application (TORA) to merge additional vessel-specific characteristics for vessels registered at the state level [[Texas Parks and Wildlife Department, 2026](#)]. Vessels weighing more than 5 net tons only require U.S. coast guard registration and do not appear in the TORA inquiry system [[United States Coast Guard, 2026](#)]. Many vessels have both, and are merged using an ID crosswalk filling roughly 79% of all vessels appearing in ticket and bid data. Unfortunately, the only vessel-specific characteristics available in both systems are year built and vessel length, though I do include addresses associated with vessel registration from TORA to approximate travel costs from home-to-bay system centroids.

I combine vessel characteristics with a historical record of auction participation, provided by TPWD, recording thousands of bids over 41 rounds from 1997 to 2024. I restrict the sample to rounds that can be merged with active landed tickets, reducing this sample to rounds 20 through 40 (years 2008 to 2023). As mentioned in the previous section, auction outcomes are dictated by TPWD’s set auction rules. Using a formula to calculate estimated market value (EMV) of the corresponding license, they ‘grant’ bids in descending order of EMV minus bid until their budget for that round would be exhausted. Bidders are allowed the opportunity to ‘accept’ this transaction or to ‘rescind’ their option to sell their license. Finally, I merge into this dataset time-varying covariates that influence vessel productivity to discipline empirical models. Average daily wind speed, measured in meters per second, and wave height, measured in meters, are taken from the closest buoy to the centroid of the five bay systems [[NOAA National Data Buoy Center, 2026](#)]. Buoys 42035 and 42020

were both offline from late September 2020 through late July 2021. Vessel-days without weather observations are excluded from estimation, so the 2020 fall and 2021 spring seasons — roughly 4,000 trips — do not enter production, choice, or structural samples. Gulf-specific diesel prices per gallon are taken from the FRED database [[Federal Reserve Bank of St. Louis, 2026](#)]. Summary statistics are reported in Table 2, organized in panels by the unit of observation.

Table 2: Summary statistics

Variable	Mean	SD	Min	Max	N
Panel A. Vessels (cross-section)					
Vessel length (ft)	37.57	11.10	15.00	70.00	1,493
Year Built	1980	10	1900	2020	1,571
Ever bid	0.47	0.50	0.00	1.00	1,593
Ever removed	0.35	0.48	0.00	1.00	1,593
<i>Active in tickets (2008–2023)</i>					
Total landings (lbs)	46,381.23	67,762.15	0.26	487,626.01	557
Bays fished	1.57	0.88	1.00	5.00	557
Years active	6.46	5.32	1.00	16.00	557
Panel B. Bay-day covariates (2008–2023)					
Vessels on water (per day)	10.11	12.80	1.00	90.00	10,885
Daily harvest (lbs)	2,153.54	4,202.19	0.15	90,024.25	10,885
Cumulative harvest (lbs)	201,345.09	285,327.13	0.00	1,654,442.99	10,885
Ex-vessel price (\$/lb)	3.04	1.99	0.62	10.51	10,885
Wind speed (m/s)	5.43	1.87	1.21	15.12	10,885
Wave height (m)	0.78	0.34	0.12	4.51	10,885
Diesel (\$/gal)	2.35	0.78	0.85	4.57	10,885
Panel C. Bay-license bids (all rounds)					
Bid (asking price, \$)	10,780.95	12,055.26	100.00	200,000.00	2,515
EMV reserve (\$)	7,230.44	1,516.39	1,880.00	10,073.08	2,515
Gap (EMV – bid, \$)	-3,550.51	12,026.77	-196,710.00	8,210.42	2,515
Granted	0.53	0.50	0.00	1.00	2,481
Removed	0.44	0.50	0.00	1.00	2,496
Rescinded	0.08	0.27	0.00	1.00	2,468

Note: Panel A covers all vessels appearing in bay tickets or bay bids; length and year built from registration records, length trimmed to 15–70 ft (80 non-commercial outliers dropped). Italicized rows describe vessels with recorded bay landings. Panel B is bay-system-by-day, in-season (May 15–Nov 30), 2008–2023, days with at least one active vessel. Panel C covers bay-license bids across all rounds; EMV computed from the TPWD matrix with era multipliers; 45 non-positive bid(s) dropped.

4 A Dynamic Model of Common-Pool Exploitation

This section develops the structural framework used to recover the continuation value of holding a shrimp license. The empirical strategy proceeds in three steps. First, I estimate harvest production and stock depletion using vessel-level trip-ticket data, isolating biological depletion from congestion. Second, I estimate vessels’ daily participation and bay-choice probabilities and use these, together with estimated state transitions, to recover structural payoff parameters and forward-simulate continuation values. Third, I merge those continuation values with the auction records to estimate how the value of remaining in the fishery affects the decision to bid. These estimated objects are then used in Section 6 to simulate the buyback’s participation and harvest-offset effects over the program’s full horizon.

4.1 Model Overview

I model the Texas inshore shrimp fishery as an infinitely repeating finite-horizon dynamic game in which heterogeneous vessels make daily participation and location decisions across multiple bay systems as a best response to rival behaviors. The framework extends [Huang and Smith \[2014\]](#), who use methods proposed by [Aguirregabiria and Mira \[2002, 2007\]](#) to estimate efficiency costs of common-pool exploitation in the North Carolina shrimp fishery using a single estuary with a binary fish/don’t-fish decision, to a spatial ‘patchy’ setting with five bay systems (Galveston, Matagorda, San Antonio, Aransas, and Corpus Christi) and endogenous location choice, following related work on common-pool resource exploitation across a patch environment [[Hotz and Miller, 1993](#), [Smith, 2005](#), [Sanchirico and Wilen, 2020](#)].

Texas inshore fisheries are open from May 15th to November 30th, with an inter-season regulatory closure from July 15th to August 15th. Because shrimp are highly fecund and migrate offshore to spawn, the cross-year recruitment overfishing problem can be ignored, and I only need to consider overfishing’ impact on biomass growth within a given year.

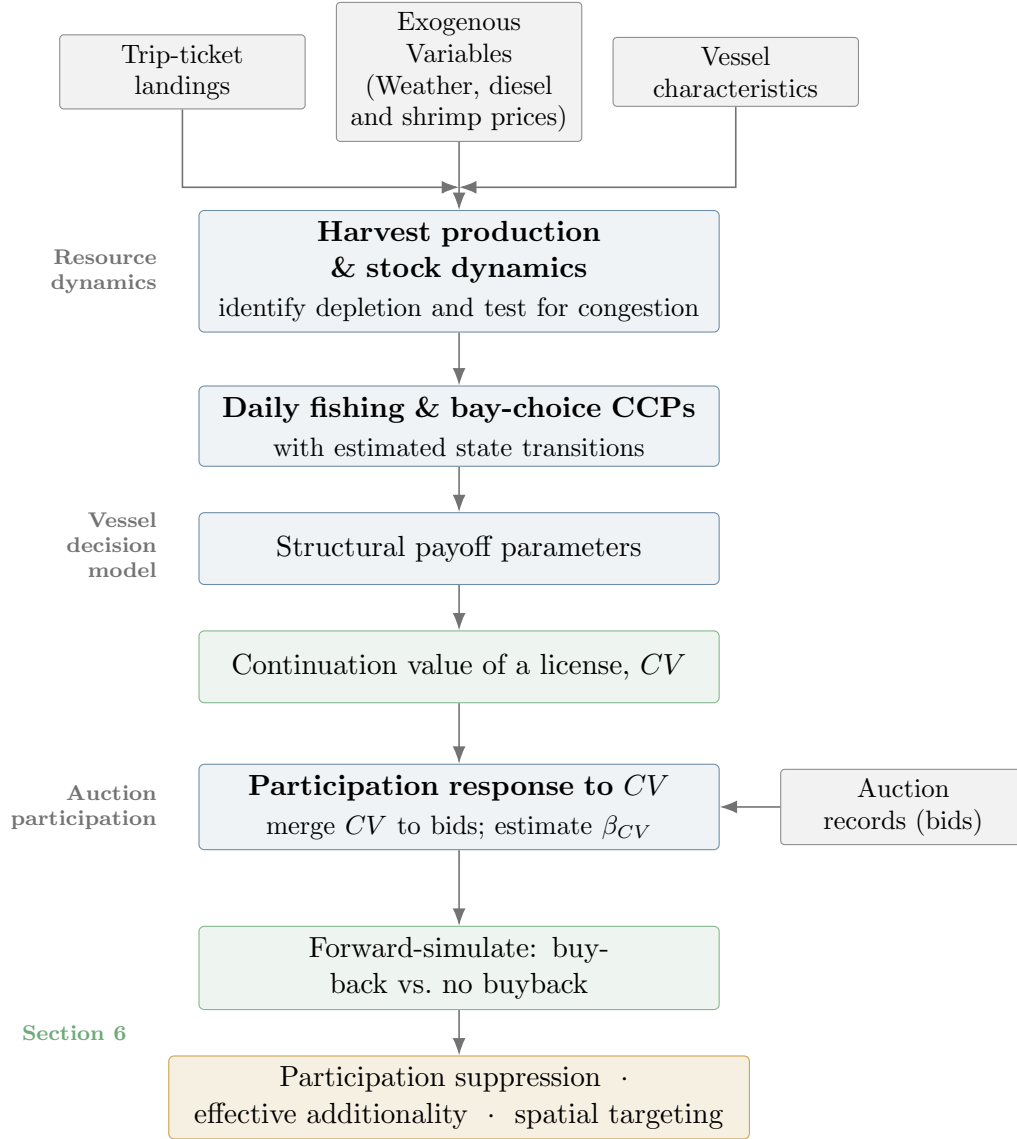


Figure 5: Overview of the structural pipeline. Raw data objects are shaded gray, estimated objects blue, simulated objects green, and results amber.

The annual cycle simplifies the stock externality problem: within-season depletion by the fleet reduces current-year stocks, which renews fully through exogenous recruitment over the winter closure. Within a season, the stock follows a single bay-specific growth path and is drawn down monotonically by cumulative harvest, and the mid-summer closure enters a shift in participation.

Each day $t = \{1, \dots, T = 200\}$ within season τ , vessel i observes a common state vector $\mathbf{s}_t$ comprising bay-specific stock indices, shrimp prices, weather conditions, diesel prices, congestion levels, and loyalty profiles across bay systems. The vessel then chooses among $J + 1$ alternatives: fish one of the J bay systems, or refrain from fishing. Vessels are forward-looking with respect to the evolution of stock, prices, and rival behavior; because any single vessel’s harvest is a negligible share of a bay’s aggregate stock, however, each treats the stock path as essentially exogenous to its own daily choice. I assume the observed fleet behavior constitutes a Markov Perfect Equilibrium (MPE).

The model is estimated in two stages. In the first stage, I recover the production technology, latent stock dynamics, conditional choice probabilities (CCPs), and state variable transition processes. In the second stage, I use the first-stage CCPs and transition estimates to simulate continuation values forward and recover the profit parameters via pseudo-maximum likelihood. The CCP approach avoids solving the full dynamic programming problem, which would be computationally intractable given the continuous state space and hundreds of strategic agents.

Two features of this environment are worth clarifying. First, the model delivers two distinct constructs with different objectives. The first includes structural payoff parameters $\boldsymbol{\theta}$ that describe preferences, such as how a vessel’s owner trades off expected revenue, stock, loyalty, travel cost, and weather in daily location choices, recovered by matching observed daily choices. The second is continuation value CV_i , the expected discounted value of holding a license and continuing to fish, which is a separate quantity generated by forward simulating the daily choice problem dictated by $\boldsymbol{\theta}$ that license-holders weigh against a buyback offer.

The second feature is to clarify that each vessel is atomistic. Because own-harvest is negligible relative to a bay’s stock, an individual’s daily choice is effectively myopic, even as the fishery remains fully dynamic in the aggregate and combined harvest depletes the common stock. The dynamic common-pool externality the buyback generates therefore does not live in any single vessel’s daily problem, but rather in the aggregate stock and, through it, in the continuation value of the license (see [A.1](#)).

4.2 Production Function and Stock Dynamics

4.2.1 Harvest Production

I assume harvest for vessel i in bay j on day t follows a restricted Cobb-Douglas production function:

$$h_{ijt} = q_i \cdot \exp(\gamma(N_{jt} - 1) + \delta'W_t) X_{jt} \exp(\varepsilon_{ijt}^h) \quad (1)$$

where q_i is vessel-specific catchability capturing unobserved fishing technology, captain skill, and experience; N_{jt} is the count of vessels fishing bay j on day t (entered as leave-one-out to avoid own-vessel contributions to congestion); W_t collects daily weather controls including wind speed $WSPD_t$ and wave height $WVHT_t$ measured at NOAA buoys [[NOAA National Data Buoy Center, 2026](#)], and summer regulatory closure indicators; X_{jt} is the latent stock index in bay j on day t ; and ε_{ijt}^h is an i.i.d. harvest shock with mean zero.

The parameter γ captures congestion or agglomeration effects. If $\gamma < 0$, additional vessels reduce individual harvest, consistent with negative congestion externalities as shrimp trawling disperses aggregations of stock. If $\gamma > 0$, additional vessels increase productivity, perhaps through information sharing that reduces search costs.

After log transformation, the production function becomes a linear regression:

$$\ln(h_{ijt}) = \ln(q_i) + \gamma(N_{jt} - 1) + \delta'W_{jt} + \ln(X_{jt}) + \varepsilon_{ijt}^h \quad (2)$$

where $\ln(q_i)$ is estimated with vessel effects.

If vessels observe bay-specific productivity shocks invisible to the researcher, a positive shock attracts more vessels, generating spurious positive correlation between vessel counts and harvest. OLS estimates of γ are therefore biased upward. [Huang and Smith \[2014\]](#) instrument total vessel count with a weekend indicator, shifting participation decisions through non-economic channels orthogonal to stock shocks. This is not applicable to my setting, however, as weekend (or other fishery-wide regulatory) shocks do not vary over j . Instead, I instrument bay-specific vessel counts using bay-specific Bartik/shift-share instruments. The instrument’s shifters — weekend and diesel — are excluded from the production function; adding them as direct controls leaves $\hat{\gamma}$ and $\hat{\phi}$ unchanged (Appendix Table [A.5](#)).

4.2.2 Latent Stock Index

The stock is not directly observed by the researcher, but is instead treated as a latent state variable observable by fishermen and endogenously influenced by their activities. I decompose the bay-specific stock index as

$$X_{jt} = A_j \cdot X_{0,\tau} \cdot G_j(t) \cdot \exp(-\phi H_{jt}^{cum}) \cdot \exp(\eta_j \mathbf{1}[\text{Harvey}]_t) \cdot \exp(\zeta_{jt}) \quad (3)$$

Where A_j represents bay system fixed effects; $X_{0,\tau}$ is the initial stock size at the beginning of year yr (captured by year fixed effects), reflecting exogenous recruitment; $G_j(t)$ is the deterministic within-season biological growth function, which is bay-specific and fixed across years; ϕ is the fishing mortality coefficient, measuring how cumulative harvest $H_{jt}^{cum} = \beta_{0j} + \sum_{s<t} h_j(s)$ depletes the stock relative to its biological baseline; β_{0j} is any accumulated harvest from January 1st through May 14th of that year; Harvey_t is an indicator for the recovery window following Hurricane Harvey’s landfall, August 25, 2017 through November 30, 2018; and ζ_{jt} is a stochastic shock to the stock index.

Cumulative harvest is treated as a predetermined state, a decision which is supported by two features of the setting. First, cumulative harvest is dated beginning-of-day, so today’s

harvest shock cannot enter it. Second, vessels are atomistic, meaning the median vessel accounts for under one percent of its bay-year harvest, and an individual vessel’s affect on it is negligible (see Appendix Table A.1). A remaining threat is persistent bay-level productivity raising both harvest and accumulation, biasing $\hat{\phi}$ toward zero. Appendix Table A.5 quantifies this with a seven-day lag of the accumulation and with a cumulated shift-share instrument, both of which give larger depletion. The record estimate is therefore conservative.

4.2.3 Within-Season Growth Function

The deterministic growth function $G_j(t)$ is modeled as bay-specific quartic polynomial in log-time:

$$\ln G_j(t) = \beta_{j1} \ln(t) + \beta_{j2} [\ln(t)]^2 + \beta_{j3} [\ln(t)]^3 + \beta_{j4} [\ln(t)]^4 \quad (4)$$

where t is days since May 15th (the approximate start of the harvest season). Note the first three coefficients vary by bay system while the fourth is pooled across bay systems due to power issues.

After estimating the production function, the composite residual $\varepsilon_{ijt}^h + \zeta_{jt}$ is recovered. Since ε_{ijt}^h is i.i.d. across vessels and ζ_{jt} is common to all vessels in bay j on day t , regressing the residual on bay-by-day fixed effects identifies ζ_{jt} .

4.3 Bay System Choice

The first-stage CCP estimation recovers the probability that vessel i chooses among $J + 1$ alternatives on day t . To accommodate the participation and location decisions jointly while allowing for correlation among the fishing alternatives, I estimated a nested logit with two nests: a fishing nest containing five bay systems and a singleton “home nest”. The deterministic utility for fishing bay j is:

$$\begin{aligned}
V_{ijt} = & \alpha_{0j} + \sum_{k=1}^3 \left[\beta_{p^k} (p_{jt}^{ma})^k + \beta_{X^k} X_{jt}^k \right] + \beta_L L_{ij,t-1} \\
& + \beta_{q,j} \ln q_i + \beta_{C,j} C_t + \beta'_{W,j} \mathbf{W}_t + \beta_{w,j} \text{wknd}_t \\
& + \sum_{k=1}^3 \beta_{t^k} (\ln t)^k + \beta_H \mathbf{1}[\text{Harvey}_t] + \tau_{\text{yr}(t)}
\end{aligned} \tag{5}$$

Coefficients subscripted by j are alternative-specific. α_{0j} is a bay-specific alternative-specific constant; p_{jt} is the shrimp price at bay j ; p_{jt}^{ma} is a 30-day weighted moving average of bay system prices; X_{jt} is the latent stock index; C_t indicates the inter-season regulatory closure; W_t includes wind speed and wave height recorded at the nearest buoy; k indexes the order of polynomials; and τ_t represent year fixed-effects. The outside option is normalized at $V_{i0t} = 0$.

Fishermen exhibit strong persistence in location choice, returning to familiar grounds even when conditions elsewhere may be more favorable [Smith, 2005, Sanchirico and Wilen, 2020]. I model this through L_{ijt} , defined as an exponentially weighted average of vessel i 's recent history of trips to bay j :

$$L_{ijt} = a_L L_{ij,t-1} + (1 - a_L) \mathbf{1}[a_{it} = j] \tag{6}$$

where a_L is the persistence of the loyalty stock, capturing the notion that vessels develop bay-specific knowledge (bottom conditions, shrimp aggregation patterns, local market relationships, etc.) that create inertia in location choice. I estimate it by the profile likelihood of the first-stage choice model over a grid of α_L (Appendix Table A.6: the likelihood peaks at $\alpha_L = 0.40$ with a 95% likelihood-ratio band of roughly 0.35-0.43, so a bay choice has a half-life of under one trip. Under the nested logit assumption with inclusive value parameter $\lambda \in (0, 1]$, the probability that vessel i chooses bay j is:

$$\Pr(j|i, t) = \frac{\exp(V_{ijt}/\lambda)}{\sum_{k \in \mathcal{J}_F} \exp(V_{ikt}/\lambda)} \cdot \frac{\exp(\lambda I_{it})}{1 + \exp(\lambda I_{it})} \quad (7)$$

Where $\mathcal{J}_F$ is the fishing nest and $I_{it} = \ln \sum_{k \in \mathcal{J}_F} \exp(V_{ikt}/\lambda)$ is the inclusive value of the fishing nest. The probability of staying home is $\Pr(0|i, t) = 1/(1 + \exp(\lambda I_{it}))$.

4.4 State Variable Transitions

With the continuation values expressed as functions of the first-stage CCPs, estimating the structural parameters requires evaluating what those continuations would be at states the vessel has not yet visited. Following [Bajari et al. \[2007\]](#), I simulate the exogenous state vector forward from each observed date t to the end of the season using estimated transition processes for each state component. Shrimp prices follow a bay-specific VAR(4):

$$\ln \mathbf{p}_{t+1} = \boldsymbol{\mu}_p + \sum_{\ell=1}^4 \boldsymbol{\Phi}_\ell \ln \mathbf{p}_{t-\ell+1} + \boldsymbol{\varepsilon}_t^p, \quad \boldsymbol{\varepsilon}_t^p \sim N(\mathbf{0}, \boldsymbol{\Sigma}_p), \quad (8)$$

where $\mathbf{p}_t$ is the J -vector of bay prices and $\varepsilon \sim N(0, \Sigma_p)$. Weather variables (wind speed and wave height) follow a VAR(3):

$$\ln \mathbf{W}_{t+1} = \boldsymbol{\mu}_W + \mathbf{A}_1 \ln \mathbf{W}_t + \mathbf{A}_2 \ln \mathbf{W}_{t-1} + \mathbf{A}_3 \ln \mathbf{W}_{t-2} + \boldsymbol{\varepsilon}_t^W, \quad \boldsymbol{\varepsilon}_t^W \sim N(\mathbf{0}, \boldsymbol{\Sigma}_W), \quad (9)$$

where $\varepsilon_t^w \sim N(0, \Sigma_w)$ and the covariance allows correlation between wind speed and wave height within a buoy. Diesel prices follow a random walk in logs,

$$\delta_{t+1} = \delta_t + \varepsilon_t^\delta, \quad \varepsilon_t^\delta \sim N(0, \sigma_\delta^2) \quad (10)$$

and stock shocks ζ_{jt} , the bay-day means of the production residual, centered within bay, follow a bay-specific AR(4):

$$\zeta_{j,t+1} = \rho_{1j} \zeta_{jt} + \rho_{2j} \zeta_{j,t-1} + \rho_{3j} \zeta_{j,t-2} + \rho_{4j} \zeta_{j,t-3} + \varepsilon_{jt}^\zeta, \quad \varepsilon_t^\zeta \sim N(\mathbf{0}, \mathbf{\Sigma}_\zeta), \quad (11)$$

These transition processes are estimated on in-season days of the 2008-2023 panel. Lag orders are chosen by information criteria and sequential likelihood-ratio tests (Appendix Table A.8); Schwarz and Hannan-Quinn select four lags for prices and three for weather. Criteria disagree for the stock shock across bays, selecting between two and six lags; I use four in every bay, which within that range everywhere.

4.5 Equilibrium and Continuation Values

4.5.1 Markov Perfect Equilibrium

Each day t , vessel i chooses action $a_{it} \in \{0, 1, \dots, J\}$ to maximize expected discounted utility over the remaining season:

$$\max_{a_{it}} \{u_i(a_{it}, \mathbf{s}_t; \theta) + \xi_{ia_{it}t} + \beta_d \mathbb{E}_t[V_i(\mathbf{s}_{t+1})]\} \quad (12)$$

where $u_i(\cdot)$ is the per-period structural payoff, s_t denotes exogenous state variables, $\xi_{ia_{it}t}$ is a private preference shock within the nest, distributed Type I Extreme Value, and β_d is the daily discount factor. The per-period structural utility for choosing bay j by vessel i is:

$$\begin{aligned} u_i(j, \mathbf{s}_t; \boldsymbol{\theta}) = & \kappa_{0j} + \alpha(p_{jt} \hat{h}_{ijt} - \bar{r}_i) \\ & + \beta_X X_{jt} + \beta_L L_{ijt} + \beta_d d_{ij} \delta_t + \beta'_W \mathbf{W}_t + \beta_\delta \delta_t + \beta_w \text{wknd}_t + \beta_C C_t + \beta_\ell \ell_i + \beta_{\ell\ell} \ell_i^2 \end{aligned} \quad (13)$$

Here, κ_{0j} represents the five alternative-specific constants of the bay system, and α represents the marginal utility of expected revenue, in the demeaned revenue of the vessel $(p_{jt} \hat{h}_{ijt} - \bar{r}_i)$ ⁶. The outside option is normalized to zero ($u_i(0, s_t; \theta) = 0$), and the structural

⁶Revenue enters this way because catchability enters revenue multiplicatively, and the revenue level is

parameter vector is $\theta = (\alpha, \beta_X, \beta_L, \beta_d, \beta_W, \beta_\delta, \beta_w, \beta_C, \beta_\ell, \beta_{\ell\ell}, \{\alpha_{0j}\}_{j=1}^5, \lambda)$. d_{ij} represents a proxy for distance, measured between a bay system's centroid and the most recently reported address associated with a license on TORA. Distance only enters as an interaction with diesel prices, δ_t , which also enters as it's own marginal effect on utility. ℓ represents vessel length⁷, $wknd_t$ is an indicator variable set to 1 for Saturdays and Sundays, and λ is a nested inclusive value. A Markov Perfect Equilibrium is a profile of strategy functions $\{\varphi_{it}^*(\mathbf{s}_t, \xi_{it})\}$ such that each vessel's strategy is a best response to the strategies of all other vessels:

$$\varphi_{it}^*(\mathbf{s}_t, \xi_{it}) = \arg \max_{a_{it}} [V_{it}^\varphi(a_{it}, \mathbf{s}_t, \xi_{it})] \quad (14)$$

where the value function accounts for the equilibrium play of all other vessels both in the current period and in all future periods. This equilibrium concept imposes that decision rules and beliefs about others' behavior are mutually consistent.

4.5.2 Forward Simulation

Under the Type I Extreme Value assumption, CCPs contain sufficient information to recover continuation values non-parametrically, avoiding backward induction entirely. Specifically, the expected value of choosing optimally prior to the realization of private shocks has the closed form:

$$V_i(\mathbf{s}_t) = \ln \sum_{j=0}^J \exp(u_i(j, \mathbf{s}_t; \theta) + \beta_d \mathbb{E}[V_i(\mathbf{s}_{t+1}) | a_{it} = j]) + \gamma_E \quad (15)$$

where γ_E is Euler's constant. Because the outside option is normalized, the log-probability

collinear with q_i ; the demeaning within the vessel identifies α from variation around each vessel's own mean rather than the confounded cross-vessel level.

⁷Length, as well as distance, are only available as the most recent recording available on the TORA and/or USCG database as of January 2025. If a license-holder changed addresses, or if the associated vessel has been capitalized since their appearance in the data, it is not observable. Bid data includes a contemporaneous record of vessel length, which I default to when available. These should be seen as noisy proxies for distance and length, respectively.

of staying home provides a nonparametric estimate of the continuation value:

$$-\ln(\hat{\sigma}_{i0t}) = \ln(1 + \exp(\lambda I_{it})) \quad (16)$$

The nested logit introduces the inclusive value parameter λ , so the continuation value of the fishing nest is:

$$CV_{it} = \lambda \ln \sum_{k \in \mathcal{J}_F} \exp(\hat{V}_{ikt}/\lambda) \quad (17)$$

To evaluate continuation values at candidate parameter values for states the vessel has not visited, I simulate the state vector forward. On each simulated day, the procedure:

1. Draws next-period state variables from the estimated transition processes (prices, weather, diesel, stock shocks).
2. Updates the stock index incorporating harvest feedback: the deterministic stock evolves via $G_j(t)$, while cumulative harvest depletes the stock through $\exp(-\phi H_{jt}^{cum})$.
3. Computes nested logit choice probabilities for all vessels given the current state, yielding predicted vessel counts by bay.
4. Updates loyalty based on realized (simulated) choices.
5. Accumulates the discounted log-sum contribution from each simulated day.

The simulation runs from each observed date t forward to the end of the season, and the discounted sum of daily log-sum values gives the continuation value. The daily discount factor β_d is set to 0.999.

4.5.3 Continuation Value Computation

The structural parameters are estimated via pseudo-maximum likelihood:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i,t} \sum_{j=0}^J \mathbf{1}[a_{it} = j] \cdot \ln \Pr(j|\mathbf{s}_t; \theta) \quad (18)$$

where the choice probabilities are constructed from the per-period payoffs $u_i(j, s_t; \theta)$, while the continuation value, common across alternatives under atomicity, cancels the conditional choice probabilities and is computed separately by forward simulation for the auction analysis. The advantage to this approach is that θ only enters through the per-period utility $u_i(\cdot)$, while continuation value is estimated nonparametrically from the CCPs and does not depend on θ , following from the atomicity of the individual vessel⁸. Because a single vessel's harvest is negligible relative to a bay's stock, the continuation value on the right-hand side of (15) is invariant to that vessel's own current choice: fishing bay j today rather than staying home leaves the aggregate stock path, and option value of continued operation, essentially unchanged. The continuation term is thus common across a vessel's alternatives on a given day, so it drops out of the conditional choice probabilities and does not confound the flow parameters θ . This is the finite-dependence logic proposed by [Arcidiacono and Miller \[2011\]](#), specialized to an atomistic common-pool setting. The one channel through which a vessel's own choice does move its future state is loyalty — fishing bay j today raises its propensity to return — but a single day's update is second-order over a season of $T = 200$ days, and I treat the continuation as invariant to it.

For the buyback analysis in subsequent sections, I convert the season-level continuation value from logsum units to money-metric continuation values using the same forward simulation with utility restricted to the expected-revenue term, divided by $\hat{\alpha}$ from Equation (13). The vessel-specific continuation value accounts for heterogeneity in catchability q_i , home port location, and loyalty profiles. Under an infinite-horizon interpretation with a 7.5 to 9% annual discount rate, the present value of the license is approximately $CV_i^{\$}/(1 - \delta)$, where

⁸This assumption appears in the data: the median vessel accounts for under one-percent (0.88%) of its bay-year harvest and the effective number of vessel per bay-year — the inverse Herfindahl index — is roughly thirteen. This holds for all bay systems, even though this paper only focuses on the five most active bay systems, where atomicity is held more strongly. No single vessel materially moves the stock it responds to (see Appendix A.1)

$\delta = 0.95$ is the annual discount rate⁹.

5 Estimation Results: Fishery Dynamics

5.1 Production Function Estimates

Table 3 reports estimates of the production function in Equation (2). Because including vessel i 's own presence in the congestion measure would mechanically generate a positive correlation between vessel counts and harvest, I use leave-one-out vessel counts ($N_{jt} - 1$) throughout. Column (1) shows the results of regressing log harvest on congestion, weather variables, cumulative season harvest, regulatory closure (July 15th to August 15th), Bay $\times$ time polynomials, and year, vessel, and bay fixed effects.

The coefficient on vessel count is positive and significant (0.017), suggesting agglomeration effects dominate congestion. However, as discussed in Section 4, this estimate is biased if unobservable stock shocks attract vessels to productive bays. While Huang and Smith [2014] use weekend closures as an exogenous instrument, this does not work in my setting as closures do not vary across bay systems and would not instrument bay-specific vessel counts in a model with bay fixed effects. I construct bay-specific Bartik/shift-share instruments from historic activity. For each vessel m and bay j , I record the historical fraction of m 's trips that go to bay j (computed while leaving own year out), with vessels that never fished a bay getting $s_{mj} = 0$. I then generate shifts, $\hat{P}_{mt}$ using a vessel-level participation logit for whether vessel m is active on day t using only common, exogenous drivers — wind speed, wave height, diesel prices, a time polynomial, and weekend, closure, and year dummies. These shifts are each vessel's predicted probability of being out on the water that day for reasons unrelated to any particular bay's conditions. Finally, I combine these to create the Bartik instruments: the share-weighted sum of rivals' predicted activity:

⁹This annual discount rate is embedded at a different horizon compared to the daily discount factor set earlier 0.999 daily discount rate used for the forward-simulation in Section 4.5.2.

$$Z_{ijt} = \sum_{m \neq i} s_{mj} \hat{P}_{mt} \quad (19)$$

Columns (2) and (3) detail the results of the 2nd and 1st stage estimation of the model using Bartik instruments for bay-specific vessel counts. After instrumenting, the coefficient associated with congestion is not statistically different from zero ($\gamma = -0.003$ (0.002)). The Wu-Hausman test rejects exogeneity of the vessel count ($p < 0.001$): the OLS agglomeration effect is the selection of vessels into productive bay-days. This is not surprising: Texas Gulf Coast bay systems are large (summing to a total of 4,363 km²) and contain multiple industries within, including crab and oyster fisheries.

5.2 Stock Dynamics

The latent stock index is recovered as the systematic component of the production residual: from equation (3), $\ln X_{jt}$ decomposes into a bay baseline A_j , an annual level $X_{0,\tau}$ (the year effect, capturing exogenous recruitment), a bay-specific within-season growth path $G_j(t)$, the cumulative-harvest depletion ϕH_{jt}^{cum} , and a serially correlated shock ζ_{jt} . Three features merit description. First, the deterministic path rises early in the season and bends downward as cumulative removals accumulate; the gap between the with- and without-harvest paths in Figure 6 is the depletion harvest creates. Second, that depletion is economically large — a one-million-pound increase in cumulative seasonal harvest lowers the stock index by roughly 18% ($\hat{\phi} = -0.204$, s.e. 0.052)¹⁰. Third, the bay-centered stock shocks are moderately persistent (per-bay AR(4), persistence 0.6-0.8); Table 4), so day-to-day deviations from the deterministic path carry over for several days rather than immediately reverting to zero. The growth-polynomial, bay, and year-effect estimates underlying the index are reported in Appendix A.2.

¹⁰Lagging the accumulation by a week gives $\hat{\phi} = -0.49$, and the cumulated shift-share instrument -0.75 (first-stage Wald 49); the highlighted estimate of -0.204 is the conservative end, see Appendix Table A.5.

Table 3: Production Function Estimates

	(1) OLS Log Harvest	(2) IV (2nd Stage) Log Harvest	(3) IV (1st Stage) Rival Vessels
<i>Congestion</i>			
Count of Vessels (minus one)	0.0170*** (0.0005)	−0.0028 (0.0020)	
Predicted rivals (shift-share IV)			0.7845*** (0.0573)
<i>Environmental controls</i>			
WSPD	0.0009 (0.0054)	0.0094 (0.0064)	0.5481** (0.1848)
WVHT	−0.0037 (0.0300)	−0.1210*** (0.0367)	−1.2234 (1.0339)
<i>Economic controls</i>			
Cumulative harvest (millions of lbs)	−0.2495*** (0.0442)	−0.2044*** (0.0518)	−1.5740 (1.6416)
Summer closure	−0.3420*** (0.0195)	−0.6595*** (0.0366)	−5.0260*** (0.8533)
Bay × time polynomial	✓	✓	✓
Harvey window × bay	✓	✓	✓
Year FE	✓	✓	✓
Vessel FE	✓	✓	✓
Bay FE	✓	✓	✓
Observations	109,986	109,986	109,986
Adj. R^2	0.356	0.324	0.641
Within R^2	0.116	0.072	0.405
RMSE	0.916	0.946	11.697
<i>IV diagnostics</i>			
First-stage F (cluster-robust Wald)		187.6	
Wu-Hausman p -value		0.000	

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors clustered by bay × day in parentheses. The endogenous regressor is the leave-one-out count of rival vessels, instrumented with a shift-share prediction of rivals. Cumulative harvest accumulates May 15–Nov 30, dated beginning-of-day, and is treated as a predetermined state variable.

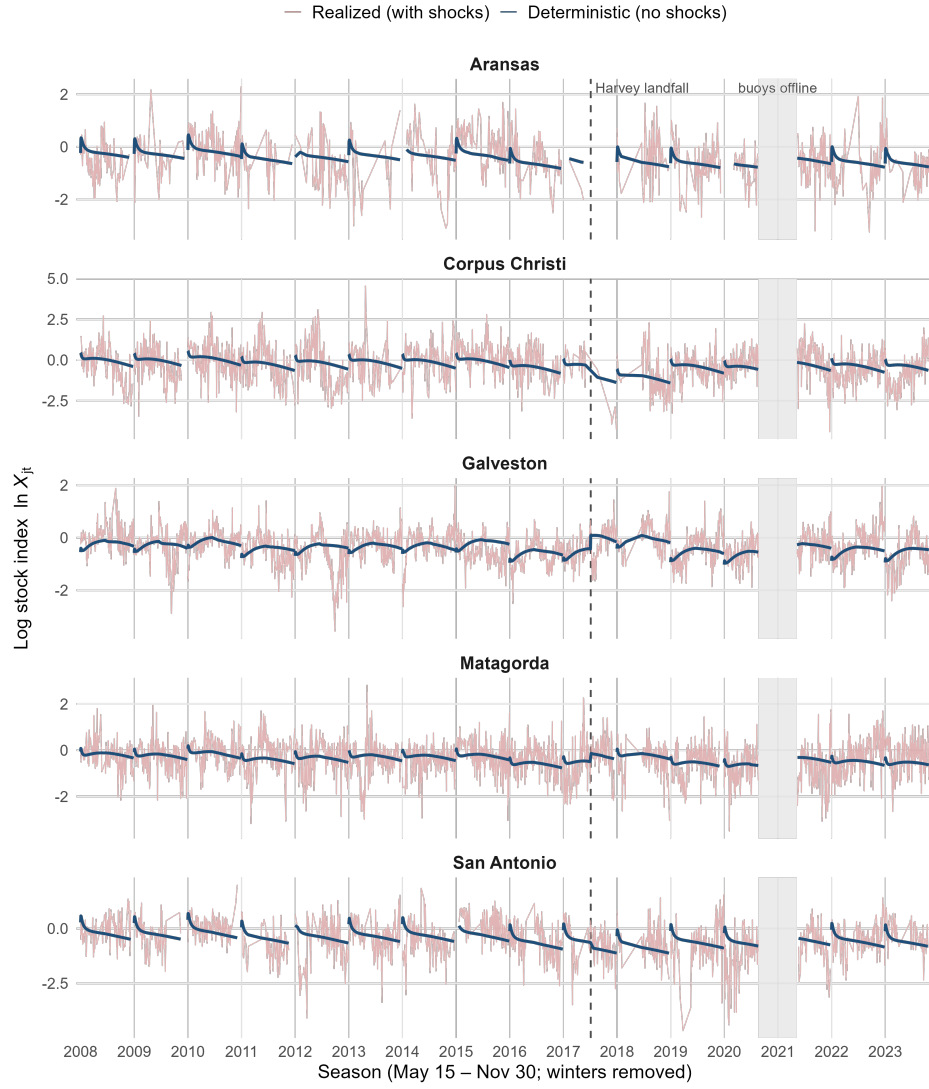


Figure 6: Logged Stock Indices by Bay System

Note: Deterministic stock indices in blue; deterministic stock overlaid in red. Stock estimated only within seasonal harvest days, May 15th to November 30th, with inter-season closure July 15th to August 15th

5.3 State Transition Fit

Following [Bajari et al. \[2007\]](#), I simulate the exogenous state vector $\mathbf{s}_t$ forward from each observed date to the end of the season, drawing each component from an estimated transition process. The components are bay-level log shrimp prices p_{jt} , wind speed $WSPD_t$ and wave height $WVHT_t$, the diesel price d_t , and the stock index X_{jt} . Dickey–Fuller tests reject the unit-root null ($p \approx 0$) for every process except diesel; I model log prices as a VAR(4) process, weather as a VAR(3) process, bay-specific shocks as an AR(4) process, and diesel prices, which retain a unit root, enter as a random walk with drift¹¹. Table 4 reports the estimates.

5.4 Stage 1 Conditional Choice Probabilities

Table 5 reports the first-stage nested logit on a 40% random draw of choice occasions¹². It’s worth mentioning that these results are not meant to carry a reduced-form interpretation, but rather to be used as a tool for simulating vessel-bay-day choices for estimating continuation values in the second stage. The inclusive-value parameter is 0.98, meaning that conditional on loyalty, stock, calendar time, and bay-specific weather and closure effects, the idiosyncratic components of bay utility are nearly independent. Loyalty is notably a dominant term ($\hat{\beta}_L = 5.1$), while wave height, wind, weekend, and mid-summer closure each lower participation in every bay.

5.5 Structural Profit Parameters

The second stage recovers the structural payoff parameters θ by pseudo-maximum likelihood, using the first-stage conditional choice probabilities and state-transition estimates to forward-simulate each vessel’s choice-specific value function. Because θ enters the per-

¹¹The orders to these processes are chosen based on information criteria and sequential likelihood-ratio tests; see Table A.8.

¹²Appendix Table A.7 shows the estimates are stable from 10% to 40%.

Table 4: State variable transition processes

	Persistence (own, $\sum$ lags)	Cross-bay ($\sum$ off-diag)	Innovation SD
<i>Bay-level log price (VAR(4))</i>			
Aransas	0.930	0.285	0.278
Corpus Christi	0.893	0.375	0.352
Galveston	0.866	0.253	0.217
Matagorda	0.758	0.522	0.434
San Antonio	0.899	0.268	0.295
<i>Weather (VAR(3))</i>			
Wind speed	0.501	0.176	0.298
Wave height	0.647	0.310	0.317
<i>Diesel (random walk with drift)</i>			
Log diesel price	1.000	—	0.0264
<i>Stock shock ζ (AR(4), by bay)</i>			
Aransas	0.768	—	0.501
Corpus Christi	0.702	—	0.757
Galveston	0.818	—	0.376
Matagorda	0.672	—	0.592
San Antonio	0.646	—	0.548

Note: Price and weather are vector autoregressions with a constant, estimated on in-season days 2008–2023; Persistence sums the own-series coefficients across lags and Cross the absolute off-diagonal coefficients across lags. Innovation SD is the square root of the diagonal of Σ . The stock shock ζ follows a per-bay AR(4) with cross-bay innovation covariance; its persistence is the sum of the four autoregressive coefficients. Companion-matrix spectral radius: 0.966 (prices), 0.642 (weather); maximum AR root modulus for ζ : 0.810–0.897. Diesel is a random walk with drift.

period payoff linearly while the continuation value is pinned down by the first-stage CCPs, the pseudo-likelihood is maximized without re-solving the dynamic program. Table 6 reports the estimates.

The parameters are economically sensible and precisely estimated. The revenue coefficient is positive — $\hat{\alpha} = 0.074$ per \$1,000 of expected revenue¹³, confirming that vessel-owners value

¹³Utility is measured in the units of the Type I extreme-value shock, so coefficients are comparable only relative to one another. The revenue coefficients supply the exchange rate: $\hat{\alpha} = 0.074$ implies a \$1,000 increase in expected revenue raises the odds of fishing that bay by $e^{0.074} - 1 \approx 8\%$, and any other coefficient

Table 5: Stage 1: Nested-Logit Location Choice

	Galveston	Aransas	Corpus Christi	Matagorda	San Antonio
<i>Panel A. Common parameters</i>					
Price level (30-day)			0.888*** (0.053)		
Price level ²			−0.227*** (0.014)		
Price level ³			0.016*** (0.001)		
Stock index X			1.417*** (0.016)		
Stock ²			−0.246*** (0.003)		
Stock ³			0.002*** (0.000)		
Loyalty L_{t-1}			5.138*** (0.073)		
$\ln t$			−0.750*** (0.066)		
$(\ln t)^2$			0.350*** (0.025)		
$(\ln t)^3$			−0.050*** (0.003)		
Harvey window			0.221*** (0.050)		
<i>Panel B. Alternative-specific parameters (relative to home)</i>					
Intercept	−7.274*** (0.149)	−7.503*** (0.292)	−4.863*** (0.161)	−6.484*** (0.174)	−7.685*** (0.361)
Wind speed	−0.044*** (0.007)	−0.019 (0.022)	−0.052* (0.021)	0.027* (0.013)	−0.026 (0.017)
Wave height	−0.270*** (0.038)	−0.235 (0.123)	−0.158 (0.113)	−0.529*** (0.071)	−0.370*** (0.095)
Weekend	−0.162*** (0.016)	−0.135** (0.052)	0.054 (0.047)	−0.215*** (0.029)	−0.394*** (0.039)
Mid-summer closure	−0.754*** (0.023)	−2.171*** (0.128)	−0.766*** (0.070)	−1.678*** (0.057)	−2.400*** (0.109)
$\ln q_i$	0.055*** (0.016)	0.030 (0.050)	−0.652*** (0.020)	−0.190*** (0.023)	0.155* (0.062)
Year shifters (fishing nest)	Yes (2008 base)				
Inclusive value (λ)	0.978*** (0.015)				
Log-likelihood	−110,667				
Choice occasions	277,200 (40% draw of 693,000)				
Vessels	471				

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Home is the reference alternative. Price level is the 30-day mean of the bay's ex-vessel price through the previous day. Price and stock enter as cubics; $\ln t$ is days since May 15. Sample: daily choice occasions, 2008–2023.

Table 6: Structural Profit Parameters (Second-Stage Estimates)

	Coefficient	Std. error
<i>Panel A. Utility parameters</i>		
Revenue (α , \$000)	0.074***	0.008
Stock index	1.583***	0.021
Loyalty	3.209***	0.060
Distance $\times$ diesel	-0.007***	0.000
Wind speed	-0.049***	0.003
Wave height	-0.324***	0.018
Diesel	-0.029***	0.005
Weekend	-0.181***	0.008
Summer closure	-0.899***	0.012
Vessel length (/10 ft)	-0.262***	0.029
Vessel length ²	0.047***	0.004
<i>Panel B. Nesting</i>		
Inclusive value (λ)	0.839***	0.017
<i>Panel C. Alternative-specific constants (rel. home)</i>		
Aransas	-5.245***	0.089
Corpus Christi	-5.416***	0.089
Galveston	-4.676***	0.087
Matagorda	-5.135***	0.088
San Antonio	-4.839***	0.087
Log-likelihood	-284,492	
Choice occasions	622,128	
Vessels	379	

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Nested-logit structural estimates on today's covariates with the vessel's observed choice set; home is the reference. Revenue is within-vessel demeaned expected revenue in thousands of dollars; loyalty uses the pre-choice stock with persistence $a_L = 0.4$. Standard errors are second-stage Hessian errors conditional on the first stage.

expected profit. Stock enters strongly and positively (1.583): a healthier bay raises the utility of fishing it. This is the structural counterpart to the stock’s role in the production function and the mechanism through which the buyback’s biological spillover ultimately operates. Loyalty is large (3.209), echoing the strong location persistence recovered in the first stage.

The cost-side terms all carry the expected signs—wind speed, wave height, the weekend indicator, and the summer closure each reduce the utility of fishing, with the closure the single largest deterrent (−0.899) — while the length terms describe a U-shaped curve with its minimum near 28 feet: above that, which covers most of the fleet, longer vessels are more likely to fish on a given day. The inclusive-value parameter is $\hat{\lambda} = 0.839$, and indicates meaningful correlation in the unobserved determinants of bay choice within the fishing nest. Fuel enters negatively both through the distance interaction and through diesel itself, so a high-diesel day lowers the utility of fishing anywhere, but more so for vessels with home addresses far from their bay of choice’s centroid.

5.6 Continuation Values and Auction Entry

The same forward simulation yields each vessel’s continuation value (CV): the expected discounted value of continued operation accumulated as the log-sum of the daily choice problem over the horizon the vessel expects to remain active, discounted. For comparison with bids I also compute a money-metric continuation value: the same forward simulation with utility restricted to the expected-revenue term, divided by $\hat{\alpha}$, which is denominated in dollars of expected revenue.

The continuation value is the object a license-holder actually forfeits. A vessel deciding whether to sell weighs the entire discounted stream of option value from continued fishing, the ability to choose when and where to fish, and to stay ashore when conditions are poor, against the auction payment. Figure 7 traces the mean continuation value across active vessels over the sample, in structural utility units (panel A) and dollars (panel B), with

divided by $\hat{\alpha}$ is that effect in thousands of dollars of expected revenue — the mid-summer closure, for example, is equivalent to a loss of $\$ \frac{0.839}{0.074} \times 1,000 \approx \$12,149$ in a day’s expected revenue.

the band showing the interquartile range. The continuation value is volatile from season to season, but shows no pronounced secular trend over the window. Its level and dispersion are descriptive context for the mechanism rather than identifying variation: the program’s effect on continuation values, and through them on participation, is recovered as the difference between the buyback and no-buyback arms of the counterfactual in Section 6.

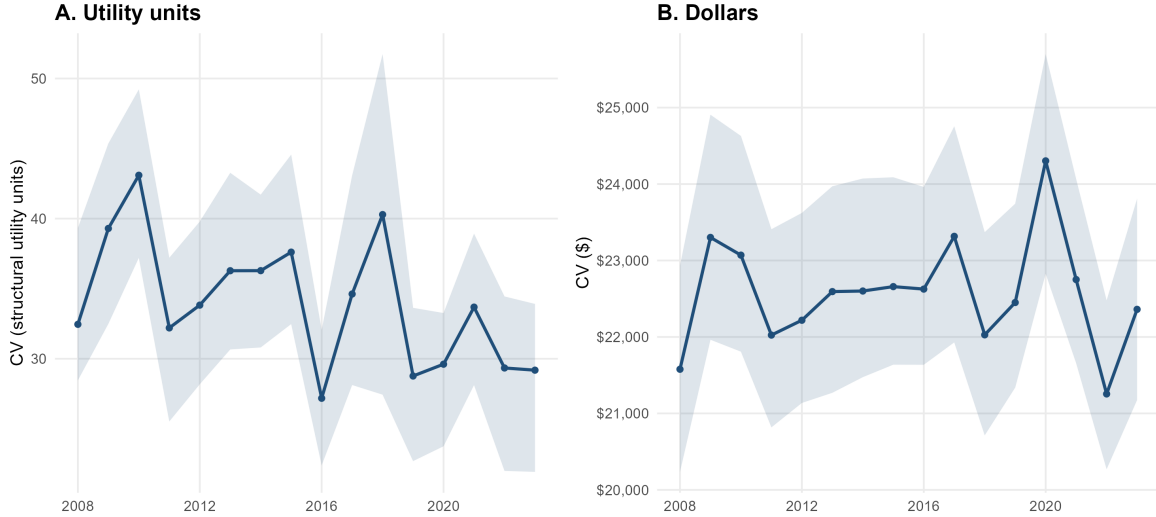


Figure 7: Mean continuation value across active vessels by season, in structural utility units (A) and in revenue-metric dollars (B); the band is the interquartile range.

Table 7 shows the estimated link between continuation value and entry into the auction, with the sign the theory predicts: a higher continuation value lowers the probability that a vessel enters a round in order to sell ($\hat{\beta}_{CV} = -0.071$)¹⁴.

I include three checks to establish that the link between continuation values and auction entry are behavioral. First, a Mundlak specification separates within-vessel from between-vessel variation in the continuation value: the within-vessel slope is -0.091 and the between-vessel slope is -0.045 and insignificant. Second, a competing-risks model separates the two ways a vessel leaves the fishery: selling at auction responds to the continuation value (-0.079), while going dormant does not ($+0.004$), and the two slopes are not poolable (Wald 10.3, $p = 0.001$). And third, splitting the simulation draws to purge simulation noise gives -0.083 .

¹⁴The risk set is every vessel-round for which a season continuation value, vessel length, and primary bay system exist ($N=4,679$ vessel-rounds), each round matched to the continuation value held before bidding.

Table 7: Auction entry and the continuation value

	(1) <i>CV</i>	(2) Expected revenue	(3) Both
<i>Variables</i>			
Continuation value CV_i	-0.0723*** (0.0214)		-0.0714** (0.0221)
Expected revenue		-0.0125 (0.0149)	-0.0024 (0.0125)
Vessel length (ft)	0.0509* (0.0227)	0.0255 (0.0186)	0.0515* (0.0229)
<i>Fixed effects</i>			
Round	Yes	Yes	Yes
Primary bay	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	4,359	4,359	4,359
Pseudo R^2	0.057	0.042	0.057

Note: Logit of the decision to bid on the continuation value and/or expected revenue, round and primary-bay fixed effects, standard errors clustered by vessel. Risk set: 4,679 vessel-rounds with a season CV, length and primary bay; each round matched to the pre-bid CV (pre-bid: month-5 - prior season; else current day-0 CV). *CV* is negative and stable across (1) and (3) while expected revenue adds nothing once *CV* is included. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

6 Counterfactual Analysis: Spillovers, Offsets, and Targeting

6.1 Overview

I move toward a counterfactual analysis to quantify the participation distortion caused by stock externalities generated by the TPWD inshore shrimp buyback auction. Due to data limitations outlined in Section 3, harvest activity is only available at the bay-year level, not vessel-bay-day level, before 2008 when a large bulk of the program's bids were placed. To construct counterfactual estimates at the program's true scale required developing a unique

approach of simulating a synthetic dataset of vessels disciplined by structural estimates of productivity, bay system choices, biological patterns, continuation values, and relationships with exogenous state variables in the 2008 to 2023 window, each using the structural estimates built in Section 5. I separately estimate the propensities of license-holders to make activity-related decisions to halt their activity outside of the auction, reactivate their license after a period of dormancy, change their bay-system preferences in response to conditions and rival activity. This spillover mediation chain is illustrated in Figure 8, with Loop A showing how license retirement undermines the program’s recruitment and Loop B showing how the program reallocates persisting effort.

This section proceeds as follows. First, I describe the building of the synthetic fleet and the process of back-casting structural objects across the full 1997 to 2023 program, with 1996 included for pre-program calibration to an unobserved stock level, in Section 6.2. I then measure the participation distortion from a biological spillover in a forward simulation, comparing arms with and without the buyback, and decomposing compositional and selection effects of the participation margin in Section 6.3. Section 6.4 measures the structural offset — that is, the own- and cross-bay recapture of removed effort and harvest in response to license retirements. Sections 6.5 uses the offset estimate to analyze counterfactual policy recommendations for specified spatial targeting and design.

6.2 The Synthetic Fleet and Full-Horizon Back-Cast

The participation footprint isolated above is measured only over the window in which the active fleet is directly observed. Trip tickets matched to individual vessels begin in 2008; before that, from the program’s 1997 inception through 2007, the landings records are compiled only at the bay level, with no vessel identifiers to place a boat in the fleet. Yet the buyback’s activity is concentrated in exactly those early years (Figure 3), when removals were largest and, because more vessels were still fishing, the stock externality they generated was plausibly strongest. A counterfactual confined to 2008–2023 therefore observes the

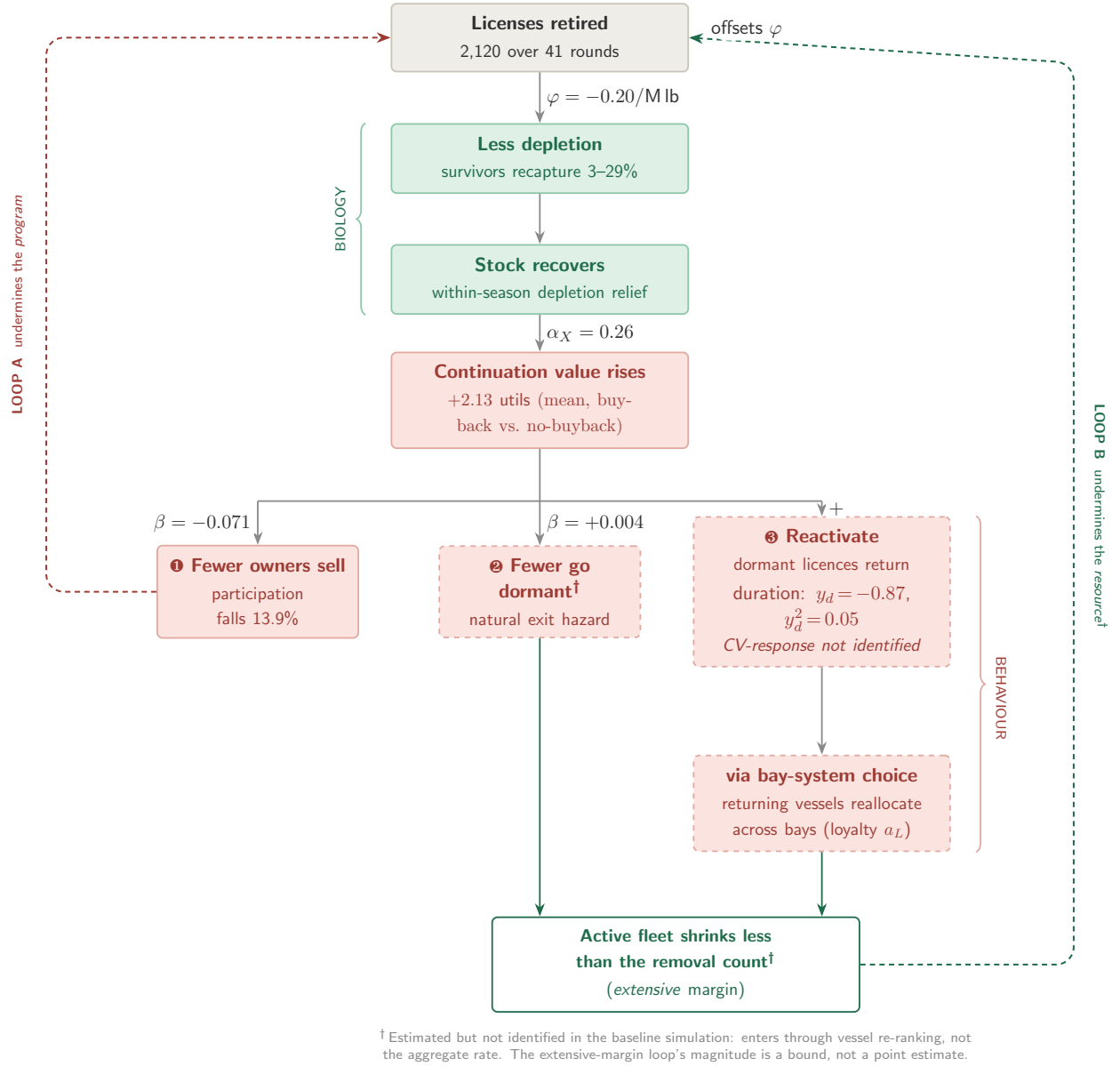


Figure 8: **Roadmap of the counterfactual mechanism.** Each round of license retirements (top) lowers cumulative depletion; survivors recapture 3–29% of the of the removed harvest (though almost none of the trawl-days), the stock recovers within the season, and continuation values rise. Higher continuation values dampen three behavioural margins—fewer owners sell, fewer vessels go dormant, and dormant vessels reactivate, reallocating across bays through the choice model. **Loop A** (participation) undermines the program by slowing removals; **Loop B** (extensive margin) offsets the biological gain. Coefficients are structural estimates from Section 5; †-marked objects are estimated but not identified in the baseline simulation.

mechanism where its force has already ebbed.

To recover the program’s full-horizon footprint, I extend the structural objects estimated in Section 5—the production function, the location-choice probabilities, and the structural preferences—across the entire 1996–2023 horizon, filling the unobserved pre-2008 period by calibration to observed moments rather than by direct estimation. Because no matched panel reaches back to 1996, the fleet that fished the early rounds is reconstructed rather than read from data: I draw a synthetic 1996 fleet from the vessels that ever appear in the license and landings records, recover the pre-2008 stock path by matching the fleet’s simulated harvest to observed 1996–2007 landings, and propagate the result forward under the estimated dynamics, anchored at the far end to the observed program-end fleet. Algorithm B.1 in Appendix B details this construction step by step—the fleet draw, the joint inversion, and the anchor calibration; the paragraphs below give its logic and, for each object, state whether it is estimated, calibrated to a moment, or left unidentified and swept.

Because Bay shrimp are short-lived and recruit annually, the stock is re-initialized each season from the estimated year effects and depleted within it. Removals propagate across years only through the reduced fleet. Endogenizing multi-year biomass persistence would matter for longer-lived stocks. Buoy weather does not reach back to 1996, so each pre-2008 season runs on a day-of-season climatology; the stock shock ζ is set to its mean of zero; prices are bay-level annual revenue per pound from the historical landings series. Each normalization is absorbed into the calibrated stock level.

The pre-2008 stock is recovered by a generalized method of moments on the treated arm. The 2008–2023 production function provides every component of the stock index except for annual levels for 1996–2007, which I parameterize as a linear trend in log stock, $X_{0,\tau} = a + b(\tau - 1996)$, plus bay-level offsets constrained to sum to zero. The moments are log in-season landings by bay-year for 1996–2007 (60) and log mean active vessels for 2008–2023 (5), with diagonal inverse-variance weights; the five bay reactivation intercepts are solved by fixed point iteration inside the objective. The estimates are $\hat{a} = 0.565$ (0.106) and

$\hat{b} = -0.0635$ (0.0153), which suggests the stock declined about six percent a year into the buyback era. The bay offsets are +0.28 for Aransas, -0.29 for San Antonio, -0.14 for Corpus Christi, +0.06 for Galveston, and +0.10 for Matagorda. Hansen’s $\mathcal{J}$ is 23.4 on 54 degrees of freedom, evidencing that the residuals show no systematic bay-level pattern (see Appendix Figure B.1). The active share of the 1996 pool (0.66), the share of removals drawn from active vessels (0.24, measured from auction records as the share of accepted 2008-2023 bids whose vessel landed shrimp in the season prior), and the loyalty-mask threshold are swept in Section 6.6. The production function, choice probabilities, structural preferences, and arrival and exit hazards are estimated (Sections 5). The 1996 stock and the reversion baseline are calibrated to observed time points. Full mechanical detail is in Appendix B.

A bidder retiring their licenses ostensibly means that their associated vessel is retired. But license holders may choose not to use their license when it’s available to them (served by the fact that about two-thirds of bidders submit zero harvest tickets in the same year), and new vessels appear regularly in the ticket data despite the 1995 moratorium on new license issuance. Two separate models are included to describe transitions of the observed fleet between dormancy and activity, detailed in Table 8. Natural exit is economically unresponsive to changes in continuation value¹⁵. Reactivation is more challenging because dormant vessels are unobservable, thus do not have an estimated continuation value. For that reason I model it as a simple reduced-form hazard rate. Appendix E compares the simulated fleet with the observed fleet over 2008–2023 on objects the calibration did not target: landings by bay-year, trawl-days per vessel, fleet size by bay, cross-bay fishing, exit and reactivation, and the characteristics of removed vessels

6.3 The Participation Distortion and its Decomposition

The buyback’s success is itself a positive spillover onto the vessels that remain, operating through a four-link chain: (i) removals lower the cumulative harvest accumulating against

¹⁵It is worth mentioning that while this model looks similar to the results in 7, the unit of observation is different, at vessel-round for the former and vessel-year for the later.

Table 8: Fleet-transition hazards (exit and reactivation)

	(1) Natural exit	(2) Reactivation
Continuation value CV_i	0.0035* (0.0017)	
Vessel length (ft)	−0.0119 (0.0096)	
Dormancy duration y_d		−0.9137*** (0.1146)
Dormancy duration ² y_d^2		0.0509*** (0.0101)
Year FE	Yes	Yes
Primary-bay FE	Yes	Yes
Observations	3,299	1,452
Pseudo R^2	0.025	0.169

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Logit hazards used in the forward simulation; standard errors clustered by vessel in parentheses. Column (1): probability that an active vessel exits for reasons other than the auction, vessel-years 2008–2023. Column (2): probability that a dormant vessel reactivates, as a function of years dormant.

the stock; (ii) lower cumulative harvest raises the stock index X_{jt} through the estimated mortality coefficient ϕ ; (iii) a higher stock raises each remaining vessel’s continuation value of holding a license; and (iv) higher expected revenue lowers the probability that a vessel enters an auction round. Links (i)–(iii) are governed by the structural production and stock dynamics of §5; link (iv) is the entry relation (Table 7).

I estimate the impacts of biological stock spillovers on auction participation using two forward simulations of the full program horizon that differ only in whether programmatic removals occur. Each arm’s fleet evolves endogenously—vessels arrive, fish, and exit under the estimated hazards, and the stock propagates through the estimated depletion—so the between-arm difference reflects spillover through an evolving fleet, the dynamic general-equilibrium form of the channel.

There are actually two separate channels that may distort participation in the two arms, which I refer to as a “matched” (behavioral) and “compositional” (selection) channel. The matched channel emerges when the same license in both arms has a different continuation value from the anticipated replenishment of stock health and reduces the license holder’s

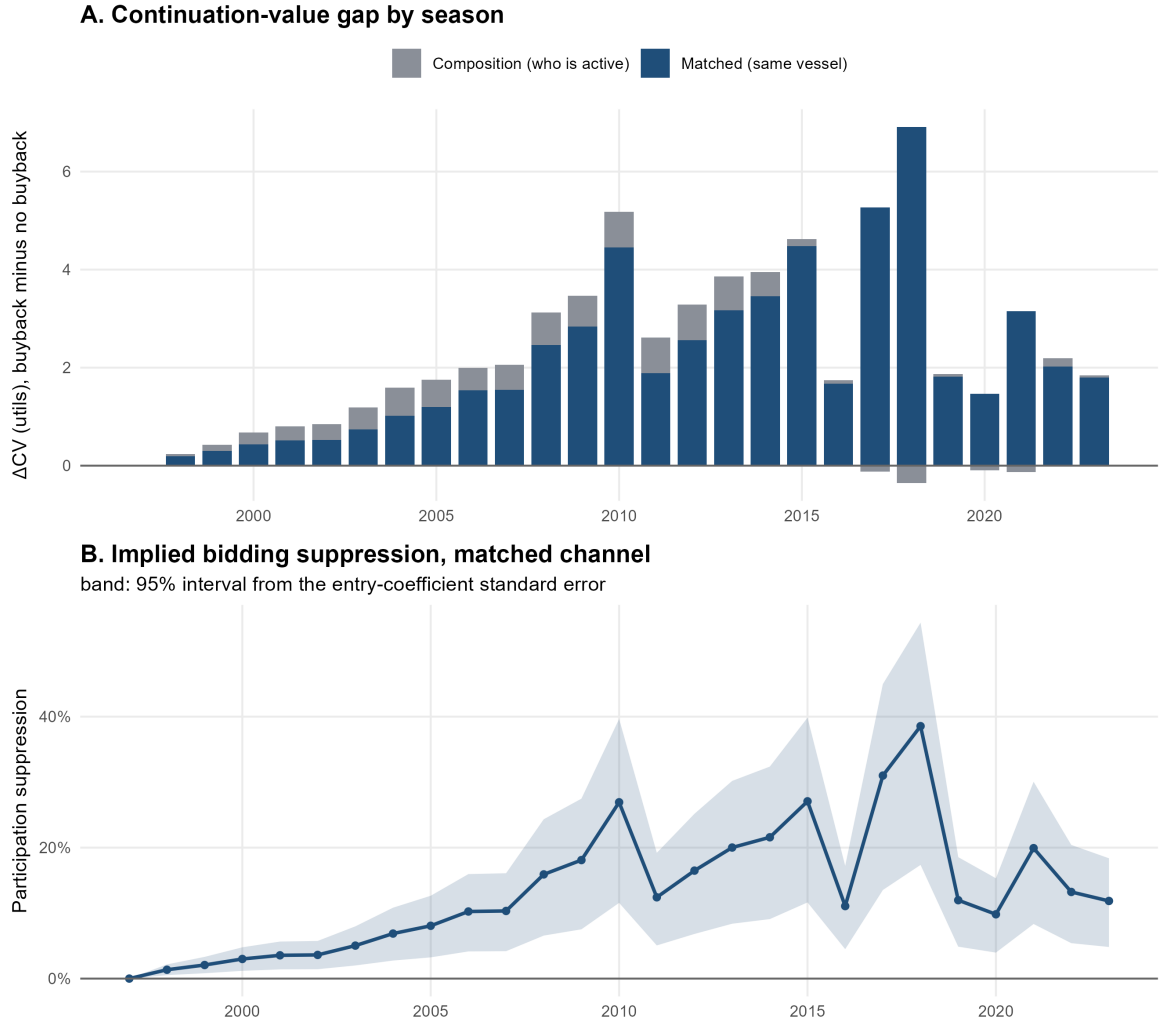


Figure 9: The participation effect by season. Panel A: mean continuation-value gap between the buyback and no-buyback arms by season, split into the same-vessel (matched) component and the composition component. Panel B: bidding suppression implied by the matched component through the entry model; the band is the 95% interval from the entry coefficient’s standard error.

propensity to bid accordingly. The compositional channel emerges from what the program intended to do — remove marginal licenses and improve the aggregate quality of active vessels, q_i , who are then less likely to bid. I decompose the two channels in a similar fashion to an Oaxaca-Blinder decomposition [Oaxaca, 1973] by matching licenses in both arms and measuring differences in propensities to bid against licensees that appear in only the no-buyback arm.

Table 9: Decomposition of the participation effect

Object	ΔCV (utils)	Entry suppression
Naive between-arm difference	2.423	15.7%
Matched (same vessel, both arms)	2.126	13.9%
Composition (who is active)	0.297	—

Note: Mean over 1997–2023 of the buyback-minus-control difference in continuation value. Matching each vessel to itself isolates its own change, which is used to calculate the entry logit ($\beta_{CV} = -0.0714$, $p_0 = 0.016$); composition is 12% of the naive gap. Across 30 synthetic-roster draws the matched suppression ranges 13.5–14.8%.

The decomposition is reported in Table 9 and illustrated in Figure 9. Comparing the same vessel operating in the buyback and no buyback arms, I find that removals raise the continuation value of an incumbent vessel by 2.13 utils on average from 1997 to 2023. This structural utility increase is about 6% of the level, equivalent to roughly \$1,400 a season on the revenue-metric scale. A further 0.29 utils of the naive between-arm gap is composition and set aside. The matched (behavioral) gain lowers the probability of bidding by 13.9%: for every six licenses that bid, one that would otherwise have bid holds out. The 95% interval is 6.0–21.5% (Appendix D.1, and it is almost entirely the sampling uncertainty of the entry coefficient. Redrawing the synthetic fleet moves the estimate by less than a point (13.5—14.8 across 30 roster seeds).

6.4 The Structural Offset

This chain is immediately suggestive of a lever that policymakers can employ to improve program efficiency — target bays (or patches) where the stock replenishment attracts less offset from those who remain. This section derives an identity for net capacity retirement contingent on endogenous patch density.

So far, I have described a demonstrable participation spillover in a program; retiring a vessel reduces capitalization in a bay, replenish stock health in that bay, license holders can harvest more shrimp at the same cost, increasing their continuation value and making

exit less appealing. But retiring a vessel does not remove its effort one-for-one from the fishery. The resource the vessel leaves behind is partly reclaimed by the vessels that remain, which now fish a marginally richer ground and expand their own effort in response. The net conservation a buyback purchases is therefore the removed effort less this offset. This subsection derives the offset from the same primitives that drive the participation channel, and I show that it saturates in the size of the surviving fleet.

Within a season, a vessel's harvest is proportional to its catchability and the local stock, $h_{ij} \propto q_i X_j$, and cumulative harvest depletes that stock, $\Delta \ln X_j = -\varphi \Delta H_j$, with the depletion elasticity governed by the Stage-1 cumulative-harvest coefficient $\beta_{\text{cum}} = 0.204$. Consider the marginal drop-one-vessel experiment underlying the counterfactual: remove an active vessel i from bay j and let the season re-solve. The removed vessel's foregone harvest $\Delta = h_{ij}$ is a positive shock to the standing stock; the S_j vessels that remain each respond by fishing the richer ground more intensively, recapturing a share of Δ . Because the recapture is itself harvest, it re-depletes the stock, so each additional surviving boat recaptures a little less than the last. Aggregating the survivors' responses gives a recapture that rises with, but saturates in, the surviving activity S_j :

$$\boxed{\text{offset}_j = \frac{\theta S_j}{1 + \theta S_j}, \quad \text{net capacity retired}_j = 1 - \text{offset}_j = \frac{1}{1 + \theta S_j}.} \quad (20)$$

The recapture is on the intensiv emargin. Table D.2 reports the offset separately for harvest and for trawl-days: survivors recapture 3-29% of a removed vessel's harvest, rising with the number of surviving vessels, but only 0-4% of its trawl-days, suggesting that, while vessel-owners catch more per day when stock is higher, they do not choose to fish more days when a neighbor leaves. A buyback in this fishery therefore retires effort almost one-for-one and harvest only partially, so the shortfall grows with fleet density because the stock gain is shared among more boats.

The single parameter θ maps the structural depletion elasticity into the strength of the

congestion-recapture feedback; estimated across bay-years it is $\theta \approx 0.246$ per million pounds, pseudo- $R^2 = 0.738$, and the saturating form fits the simulated drop-one responses well (pseudo- $R^2 \approx 0.738$), materially better than a linear recapture, which overshoots in thick bays. Equation (20) carries the paper’s central tension in one line: where the surviving fleet is large, most of a removal is offset and net conservation is small; where the fleet is thin, a removal sticks. Figure 10 plots the simulated recapture against surviving activity by bay, with the fitted curve overlaid.

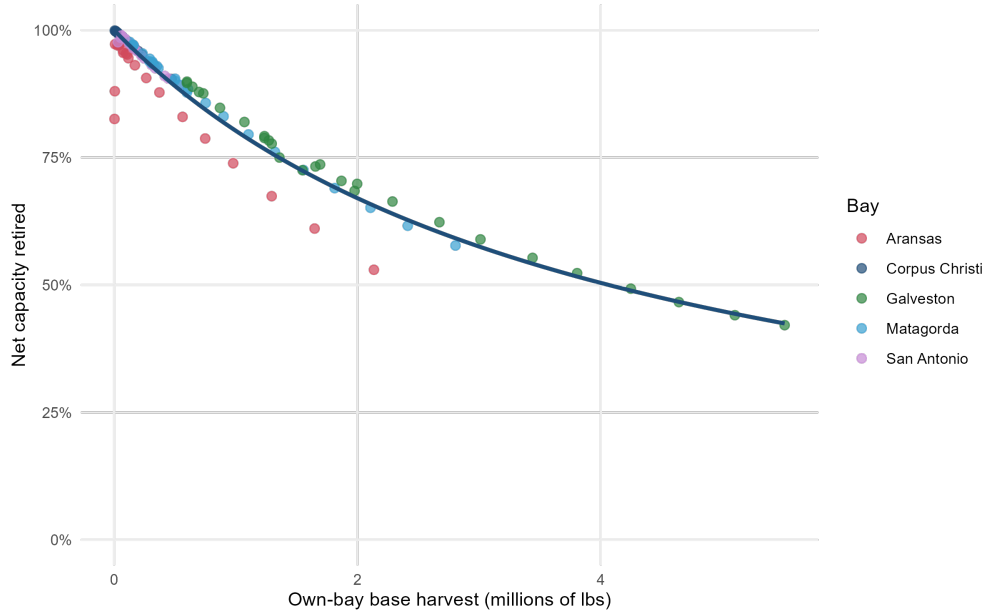


Figure 10: Net capacity retired versus the size of the surviving fleet, one point per bay-year (70 bay-years, every other year 1997–2023). The solid curve represents the structural identity in (20)

In principle the effort freed by a removal could be recaptured anywhere the removed vessel’s rivals fish, so the general object is a mobility matrix M whose (j, k) entry is the share of bay- j removals recaptured in bay k . Decomposing each removal’s recapture by where it lands based on own-bay (local congestion relief) versus cross-bay (spatial substitution) shows that the cross-bay component is small except in Aransas, whose vessels fish the neighboring bay: there the offset is entirely cross-bay. This is what the fishery’s loyalty structure would predict—vessels fish close to home, so the effort a removal frees is reclaimed mostly by boats

in the same bay.

Table 10: Conservation per marginal removal, by bay

Bay	Retention share	Average vessel (lbs)	Marginal seller (lbs)	Seller / average	Retired per avg. vessel	Retired per marginal seller
San Antonio	0.96	4,526	3,274	0.72	4,295	3,114
Galveston	0.71	10,496	2,999	0.29	7,185	2,181
Matagorda	0.86	7,984	1,920	0.24	6,562	1,642
Aransas	0.87	4,412	1,878	0.43	3,510	1,552
Corpus Christi	0.98	2,632	506	0.19	2,537	493

Note: Pounds of harvest retired per removal = retention share (matched: the same vessel, in this bay) $\times$ the removed vessel’s own harvest (composition: who sells here). The marginal seller is the lowest-continuation-value active vessel in the bay. Means over bay-years 1997–2023.

Pounds retired per removal factor into two objects: the share of the removed harvest that stays retired (matched), and how much the vessel that sells was catching (composition). Holding the vessel fixed, removal from a dense bay retires less effort. A Galveston removal, for instance, retains 71% of harvest compared to 86-98% elsewhere. However, the marginal seller in a large fleet is a small boat: Galveston’s sells 29% of an average Galveston vessel’s harvest, Corpus Christi 19%, and San Antonio 72%. The product ranks San Antonio first (3,114 lbs per removal), Galveston second (2,181), and the remaining bays at 493-1,642; see Table 10).

6.5 The Cost-Effectiveness of Removal Rules

A buyback’s job is to retire fishing effort, so a natural way to grade program efficiency is removals per dollar spent. One way to do this would be by crediting each retired license with its own effort—cost per license, or gross days removed. That accounting misses two things this paper has identified: The first is the dynamic offset of Section 6.4: the boats that remain recapture part of the freed stock, so net effort falls by less than the removed vessel’s own effort. The second is additionality: a share of the licenses retired belonged to vessels that were barely fishing, so buying them removed little real effort to begin with. Pricing both turns cost-effectiveness from an accounting number into a design problem with two

administrable levers. I measure it as net trawl-days avoided per license retired, holding the budget fixed, and benchmark each rule against the naive per-license credit a static evaluation would assume (Figure 11).

Table 11: Cost-effectiveness of removal rules

Rule	Licences retired	Net trawl-days avoided	Per licence	Share of naive credit
Program as run	1,102	187,922	171	18%
<i>Activity floor on the active pool only</i>				
25 prior-season days	1,102	188,172	171	18%
50 prior-season days	1,004	179,575	179	18%
<i>Screen: dormant licences excluded</i>				
active licences only	1,042	319,832	307	32%
active licences, 25-day floor	1,029	318,975	310	32%
<i>All removals from one bay</i>				
Galveston	766	219,095	286	29%
Matagorda	334	81,742	245	25%
San Antonio	74	33,286	450	46%
Corpus Christi	107	29,095	272	28%
Aransas	339	27,545	81	8%

Note: Net trawl-days avoided = gated trawl-days in the no-buyback arm minus the rule’s arm, 1997–2023, divided by licences retired under the rule. The naive credit of the median number of trawl days over program duration (974 days) is the mean trawl-days of an active vessel in the no-buyback fleet.

The program as run reduces trawl-days by 171 per retired license, only 18% of the naive credit of 974 days a static evaluation would assign (i.e. the median number of trawl days for each vessel over the program’s duration). Three wedges separate the program as run from the naive credit. The first is additionality at purchase: three-quarters of the licenses TPWD bought belonged to vessels with no prior-season activity. Restricting purchases to active licenses raises trawl-days retired per license by 80% to 307 (see Table 11). An activity floor on top of it, 25 or 50 prior-season days, add nothing, so the rule is not exploitable by vessels fishing a single day in the season to achieve eligibility (the marginal active seller already fishes 40 days). The second wedge is attrition after purchase: a retired vessel only retires

the days it would actually have fished before leaving on its own, and with an annual exit hazard near 16% that is only a few seasons. The third is the effort offset, which, as shown in Section 6.4, is small in trawl-days. Bay targeting improves the marginal license — San Antonio retires 450 days per license, Galveston 286, Aransas 81 — but only Galveston has the fleet to absorb the program’s removal count. Auction prices are held fixed throughout; the cost per license of an active-only screen in an empirical question the biddata can answer and is left for future work¹⁶.

6.6 Robustness

Figure 11 varies each assumption the simulation takes, holding the calibration fixed. Only two of these assumptions have a meaningful effect on participation suppression. The share of removals drawn from active vessels (assumed 0.24 in the main result, estimated from observed auction data) moves the headline from 11.2% (0.10) to 16.4% (0.45). The real active sellers fished fewer days than the simulation’s active removals, so the observed-activity end of this range is a lower bound. Loyalty shifts the result substantially: scaling the loyalty coefficient by 0.6 to 1.2 moves the effect from 5% to 24%, because loyal vessels stay in the bay that improved and compound the gain, while mobile vessels arbitrage it away, though the coefficient’s own uncertainty is a few percent. A loyalty mask limiting each vessel owner’s choice set changes the estimate only when removed entirely (16.5% where mask = 0); the estimated exit hazard’s sensitivity to the continuation value and its reference point do not move it; removal leakage of 24% lowers it to 12.2%; And an eligibility floor on prior-season days changes nothing below 50 days. However, across any (reasonable) change in baseline assumptions, the participation suppression remains positive and statistically significant.

¹⁶San Antonio’s ranking as the bay system with the most efficient per-license trawl-day reduction can seemingly contradict the fact that Matagorda and Aransas are less dense. Appendix Figure D.1 illustrates that while this is true, San Antonio also happened to have had the highest trawl-days per license initially, so while it is middle of the road at the intensive margin it dominated at the extensive margin.

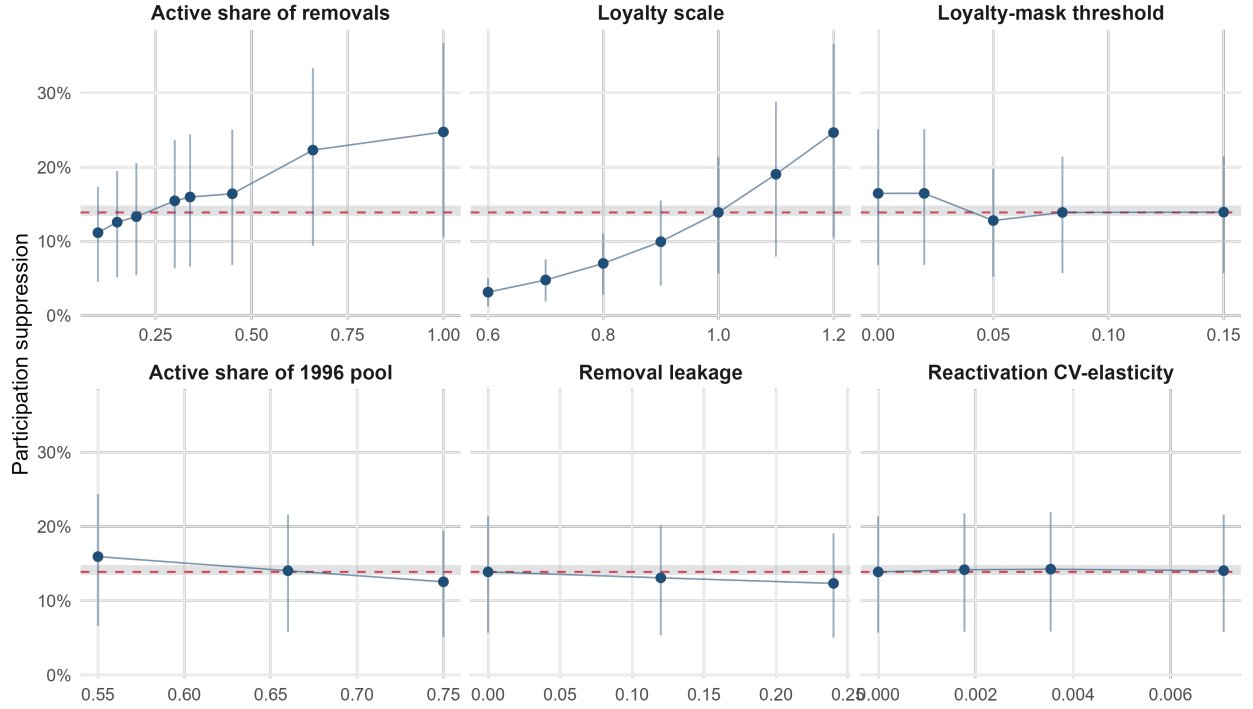


Figure 11: Sensitivity of the participation effect to simulation assumptions. Each panel varies one assumption of the counterfactual simulation with the calibration held fixed and re-solving the reactivation intercepts where the fleet composition changes; the horizontal line is the record estimate (13.9%). Error bars are 95% intervals from the entry coefficient's standard error at each point's matched ΔCV . Assumptions: share of the 1996 license pool that is active; share of removals drawn from active vessels (record 0.24, from the auction records); the loyalty coefficient scaled by 0.6–1.2; the loyalty-mask threshold on the choice set (0 = no mask); leakage of removed vessels back into the fleet; the exit hazard's sensitivity to the continuation value in units of the estimated coefficient; and the reference point for the exit hazard.

7 Conclusion

This paper structurally identifies a feedback through which the success of a buyback can weaken its own future participation: removals improve the resource, raise the continuation value of remaining agents, and make subsequent exit less attractive. The counterfactual implies that stock recovery suppressed bidding among active license-holders by about 13.9%, roughly one holdout for every six who bid. These missing bids come from licenses associated with active fishing effort, so the distortion reduces retirements that deliver conservation. The effect is negligible in the early rounds and grows as removals accumulate, making it a feature of repeated programs: a one-shot buyback would avoid the later rounds in which realized recovery suppresses bids, although forward-looking owners could still hold out in anticipation of it. Furthermore, because retired vessels would have exited within a few seasons, the effort a buyback retires is bounded by the fleet’s own attrition, which a naive per-license accounting would have ignored.

The results also have direct implications for buyback design. The value of a retirement is heterogeneous: even before a purchase occurs, local resource density affects how much conservation that retirement ultimately delivers, because remaining users recapture part of the freed resource. Screening for recent activity is one obvious response to non-additional purchases, but such rules may themselves induce gaming—for example, an otherwise inactive license-holder could fish minimally to preserve eligibility. A more robust design principle is to target patches where offsets are low and the marginal seller is still a substantial producer, and, more generally, to condition payments on expected net conservation instead of gross retirement. The same logic extends beyond fisheries wherever retirements alter the returns to remaining users: groundwater rights may be more valuable to retire where connected pumping responses are limited, while land-retirement programs may deliver more conservation where displaced production has fewer margins for reallocation.

Some limitations merit mention. First, vessel-level activity is unavailable before 2008, so the early program fleet is reconstructed rather than directly observed. The resulting back-

cast is disciplined by observed harvest and late-period fleet moments, but inevitably relies more heavily on model structure than the 2008–2023 estimation window. Transitions between dormancy and active fishing are particularly difficult to recover because dormant vessels are, by definition, unobserved while inactive. Second, the simulated fleet exits more slowly than the observed fleet after 2017 and under-represents in-and-out fishing (Appendix E); both leave more survivors to absorb the stock gain and bias the participation effect toward zero.

The counterfactual simulations in this paper evaluate the cost-effectiveness of alternative removal rules while holding the underlying auction mechanism fixed. They are therefore deliberately not welfare optima. A full welfare analysis would need to value conservation benefits against fisher surplus and allow both participation and bids to respond endogenously to changes in program rules. A related extension is to design transfers that internalize the externality a participant imposes on others. In the spirit of VCG mechanisms, such payments could in principle align private incentives more closely with the social value of retirement, although doing so may create substantial budget-balance and implementation challenges. Both exercises require a model of endogenous entry and strategic bidding and therefore constitute a distinct mechanism-design problem beyond the scope of the present paper. The structural objects recovered here provide much of the empirical foundation for that extension. For the present analysis, the contribution is narrower: to identify how much conservation alternative designs deliver per dollar once non-additional purchases and endogenous effort recapture are accounted for.

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A Supplementary Estimation Results

Table A.1: Vessel concentration and the atomicity assumption

Bay system	Median vessel share (%)	99th pctl share (%)	Effective # vessels	Median # vessels	Top-vessel share (%)
<i>Modeled bay systems</i>					
Galveston	0.52	4.8	55.9	118	4.3
Matagorda	0.90	13.4	25.0	69	8.4
San Antonio	1.70	11.5	20.8	40	8.7
Corpus Christi	2.13	18.9	12.5	28	16.3
Aransas	1.58	32.2	8.0	24	22.8
<i>Other Texas bay systems</i>					
Upper Laguna Madre	11.66	100.0	1.9	4	61.0
Lower Laguna Madre	56.89	100.0	1.4	2	85.2
Sabine Lake	94.98	100.0	1.0	1	100.0
Other	100.00	100.0	1.0	1	100.0
All bays	0.88	29.5	13.2	30	15.0

Note: Each vessel's share is its bay-shrimp landings in a bay-system-year divided by that bay-year's total. Effective vessels is the inverse Herfindahl index; the last three columns are medians across years. The lower block is outside the estimation sample.

Table A.2: Deterministic components of the stock index

Term	Aransas	Corpus Christi	Galveston	Matagorda	San Antonio
$\ln t$	1.213 (0.192)	0.168 (0.234)	0.461 (0.162)	0.467 (0.168)	0.826 (0.244)
$(\ln t)^2$	-0.811 (0.119)	-0.376 (0.124)	-0.453 (0.104)	-0.504 (0.107)	-0.697 (0.127)
$(\ln t)^3$	0.186 (0.029)	0.136 (0.028)	0.149 (0.027)	0.154 (0.028)	0.174 (0.029)
Harvey window η_j	0.045 (0.134)	-0.606 (0.152)	0.514 (0.062)	0.364 (0.067)	-0.240 (0.078)
$\ln A_j$ (bay FE, incl. $\bar{\zeta}_j$)	-0.232	0.458	-0.535	-0.051	0.281

Note: From the production regression of Table 3. Pooled $(\ln t)^4 = -0.0146$ (0.0025); depletion $\hat{\phi} = -0.2044$ (0.0518) per million lb; mid-summer closure -0.660; year effects $\ln X_{0,\tau}$ range -0.486 to 0.115 (2008 base). Standard errors in parentheses.

Table A.3: Bay-shrimp inverse demand: OLS and weather-IV

	OLS	IV (weather)
Log landings (flexibility)	-0.103*** (0.011)	-0.100* (0.045)
Implied demand elasticity	-9.7	-10.0
First-stage F		20.4
Observations (bay-days)	10,884	10,884
Clusters (year-month)	106	106

Note:

Log bay-day ex-vessel price on log bay-day landings with bay and year-month fixed effects; SEs clustered by year-month. Instruments: wind speed and wave height. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.4: Auction entry: continuation-value coefficients across specifications

Specification	$\hat{\beta}_{CV}$	S.E.
<i>Panel A. Baseline</i>		
Continuation value only	−0.0723***	(0.0214)
Continuation value and expected revenue (record)	−0.0714***	(0.0221)
expected revenue in the same model	−0.0024	(0.0125)
<i>Panel B. Sample and unobserved heterogeneity</i>		
Rounds before 2016 only	−0.0595**	(0.0235)
Correlated random effects: within-vessel CV	−0.0905***	(0.0232)
Correlated random effects: vessel-mean CV	−0.0450	(0.0380)
Vessel fixed effects (conditional logit)	−0.1177***	(0.0384)
Frailty hazard with vessel random effect	−0.0668**	(0.0331)
<i>Panel C. Identification</i>		
Stock-driven CV component only, as regressor	0.0011	(0.0048)
Control function: CV instrumented by stock component	−0.0730***	(0.0224)
Split-draw instrument (simulation noise)	−0.0832***	(0.0261)
<i>Panel D. Competing risks (multinomial: stay, dormant, sold)</i>		
Sold at auction	−0.0793***	(0.0258)
Went dormant	0.0035*	(0.0017)
<i>Panel E. Reservation value (\$ thousands; round trend)</i>		
EMV payment	2.630	(1.648)
Present value of continuation value	−0.0001	(0.0000)

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Logit of bidding on the season continuation value with round and primary-bay fixed effects unless stated; standard errors clustered by vessel. Panel D: Wald test of equal slopes on the two exits, 10.25 ($p = 0.001$). Panel E: Wald test of $\beta_{EMV} + \beta_{CV} = 0$, 2.55 ($p = 0.11$); licence age excluded from the entry equation.

Table A.5: Production function: robustness of the depletion coefficient

	Rival vessels γ	Cumulative harvest ϕ	First-stage Wald
(1) Record: bay, vessel and year FE; Harvey window	−0.0028 (0.0020)	−0.2044*** (0.0518)	187.6
(2) Bay $\times$ year fixed effects	0.0103*** (0.0027)	−0.3245*** (0.0800)	94.4
(3) Cumulative harvest lagged seven days	−0.0017 (0.0018)	−0.4901*** (0.0485)	216.2
(4) Instrument shifters (weekend, diesel) added as controls	−0.0018 (0.0019)	−0.2122*** (0.0510)	206.3
(5) No Harvey window	−0.0027 (0.0020)	−0.2141*** (0.0514)	189.7
(6) OLS, no instrument	0.0170*** (0.0005)	−0.2446*** (0.0442)	—
(7) Cumulative harvest also instrumented (cumulated shift-share)	0.0006 (0.0018)	−0.7487*** (0.0666)	104.1 48.9

Notes: Dependent variable is log vessel-day harvest; 109,986 vessel-days, 2008–2023. Each row is the record specification of Table 3 with one change. Standard errors clustered by bay $\times$ day in parentheses; $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. First-stage Wald is for the rival-vessel count; row (7) reports it for the vessel count (upper) and for cumulative harvest (lower). Row (2) absorbs each bay’s annual level and identifies ϕ from within-season variation alone; rows (3) and (7) purge persistent bay-level productivity from the accumulation and give larger depletion, so the record estimate is a conservative boundary.

Table A.6: Profile likelihood for the loyalty persistence a_L

a_L	Log-likelihood	ΔLL	$\hat{\beta}_L$	$\hat{\lambda}$	In 95% band
0.30	−55,152.0	−7.7	5.490	1.036	
0.35	−55,145.9	−1.6	5.224	0.978	yes
0.40	−55,144.3	0.0	5.009	0.929	yes
0.45	−55,148.9	−4.6	4.834	0.890	
0.50	−55,161.8	−17.5	4.689	0.858	
0.55	−55,185.7	−41.4	4.564	0.831	
0.60	−55,223.8	−79.5	4.450	0.806	

Notes: CCP of record refit on a fixed 20% draw for each a_L ; band from the likelihood-ratio criterion $2\Delta LL \geq -\chi^2_{1,0.95}$.

Table A.7: CCP estimates by draw fraction

Draw	Occasions	Vessels	LL	λ	Price level	Stock	Loyalty	Minutes
10%	69,300	471	−27534.7	0.866	0.844	1.379	4.645	4.4
25%	173,250	471	−69210.6	0.850	0.883	1.488	4.563	9.0
40%	277,200	471	−110761.4	0.869	0.882	1.363	4.651	10.3

Table A.8: Lag-order selection for the state-transition processes

Process	Dim.	AIC	HQ	SC (BIC)	Used in model
Panel A. Vector autoregressions					
Ex-vessel prices (5 bays)	5	9	4	4	4
Offshore weather (WSPD, WVHT)	2	10	3	3	3
Panel B. Stock-shock AR by bay (ζ_{jt})					
Aransas	1	5	—	2	4
Corpus Christi	1	6	—	4	4
Galveston	1	6	—	6	4
Matagorda	1	6	—	6	4
San Antonio	1	6	—	4	4
Panel C. Diesel (random walk in logs)					
Diesel, $\Delta \ln$	1	—	—	—	0 (RW)

Note: Information-criterion-minimizing lag order for each transition process; “Used in model” is the order in the forward simulation. Panel A: reduced-form VARs with a constant (SC and HQ agree on 4 and 3; AIC/FPE select higher, as they do in daily data). Sequential LR tests: the price VAR’s likelihood gain falls from 257 (lag 4) to 57 (lag 5); the weather VAR’s from 66 (lag 3) to 14 (lag 4). Panel B: per-bay AR(p) on the bay-centered stock shock ζ_{jt} , consecutive days only, searched over one to six lags; AIC selects five or six in every bay and BIC between two and six. The simulation uses four in every bay. Panel C: ADF fails to reject a unit root in log diesel (ADF = -2.52 , $p = 0.36$) and rejects after differencing (ADF = -43.94 , $p = 0.01$).

B Synthetic-Fleet Construction and Full-Horizon Back-Cast

This appendix gives the mechanical detail behind the back-cast summarized in Section 6.2: the 1996 fleet draw, the joint inversion that recovers the initial per-bay stock, the reversion-baseline calibration, and the season simulator and forward loop that the two policy arms share.

Algorithm B.1 Synthetic-fleet construction and full-horizon back-cast

Inputs: pool $\mathcal{P} = \{(q_i, \text{len}_i, b_i)\}$ (≥ 20 trips); active share σ_a ; size N ;
estimated $\Theta = \{\phi, G_j, \theta, a_L, \lambda, \text{hazards}\}$;
targets $H_{j,1996}^{\text{obs}}$ (1996 landings), F_j^{obs} (2018–2023 fleet).
Output: fleet $\mathcal{F}_{1996}$, stock $\mathbf{X}_{1996}$, reversion baseline $\{\bar{X}_j\}$.

Phase A. Draw the 1996 active fleet.

- 1: **for** each bay $j = 1, \dots, 5$ **do**
- 2: $s_j \leftarrow$ 1996 active share from historical landings, net of productivity
- 3: draw $n_j = \lfloor \sigma_a N s_j \rfloor$ vessels from $\mathcal{P}$, bay j , with replacement
- 4: **end for**
- 5: keep q_i ; impute length; inherit distances; assign loyalty a_L ; inherit revenue/lb

Phase B. Calibrate the 1996–2007 stock levels by simulated method of moments.

- 6: parameterize $\ln X_{0,\tau} = a + b(\tau - 1996) + \delta_j$ for $\tau = 1996, \dots, 2007$, with $\sum_j \delta_j = 0$
 - 7: moments: log in-season landings by bay-year 1996–2007 (60); log mean active vessels by bay 2008–2023 (5)
 - 8: weights: diagonal, inverse variance of each observed series
 - 9: outer: Nelder–Mead over (a, b) ; inner: solve the five reactivation intercepts $\{r_j\}$ by fixed point so 2008–2023 fleet sizes match
 - 10: offsets $\{\delta_j\}$ set to zero each bay’s mean landings residual at $(\hat{a}, \hat{b})$; intercepts re-solved
 - 11: inference: numerical Jacobian, sandwich covariance; report $J = g'Wg$
 - 12: **return** $\mathcal{F}_{1996}$, $(\hat{a}, \hat{b}, \hat{\delta})$, $\{\hat{r}_j\}$
-

Algorithm B.2 Season simulator and forward back-cast (helpers for Algorithm B.1)

```

1: procedure SIMSEASON( $\mathcal{F}, \mathbf{X}_0; W, \zeta, p$ )
2:    $H_j^{\text{cum}} \leftarrow 0, H_j \leftarrow 0$  for all  $j$ 
3:   for each day  $t$  in the season do
4:      $X_{jt} \leftarrow G_j(t) - \phi H_j^{\text{cum}} + \dots$  ▷ compute  $X_{jt}$ .det
5:     each vessel picks a bay (or home) by the nested-logit CCP given  $X_{jt}, W, p$ 
6:     realize  $h_{ijt}$ ; add to  $H_j^{\text{cum}}$  and  $H_j$ 
7:   end for
8:   return  $\{H_j\}$  and realized activity
9: end procedure

10: procedure FORWARDSIM( $\text{arm}; \mathcal{F}_{1996}, \mathbf{X}_{1996}, \bar{X}, \Theta$ ; schedule  $R$ )
11:    $\mathcal{F} \leftarrow \mathcal{F}_{1996}, \mathbf{X} \leftarrow \mathbf{X}_{1996}$ 
12:   for  $\tau = 1996, \dots, 2023$  do
13:      $\{H_j\}, \text{activity} \leftarrow \text{SIMSEASON}(\mathcal{F}, \mathbf{X}; W_\tau, \zeta_\tau, p_\tau)$ 
14:     if  $\text{arm} = \text{treated}$  and  $\tau \in R$  then
15:       retire licenses per schedule, ranked by entry propensity
16:     end if
17:      $\mathcal{F} \leftarrow \text{CHURN}(\mathcal{F}, \text{activity}, \Theta)$  ▷ forward_churn
18:     record participation, stock,  $CV$  for  $\tau$ 
19:   end for
20:   return participation, stock, fleet,  $CV$  paths
21: end procedure

```

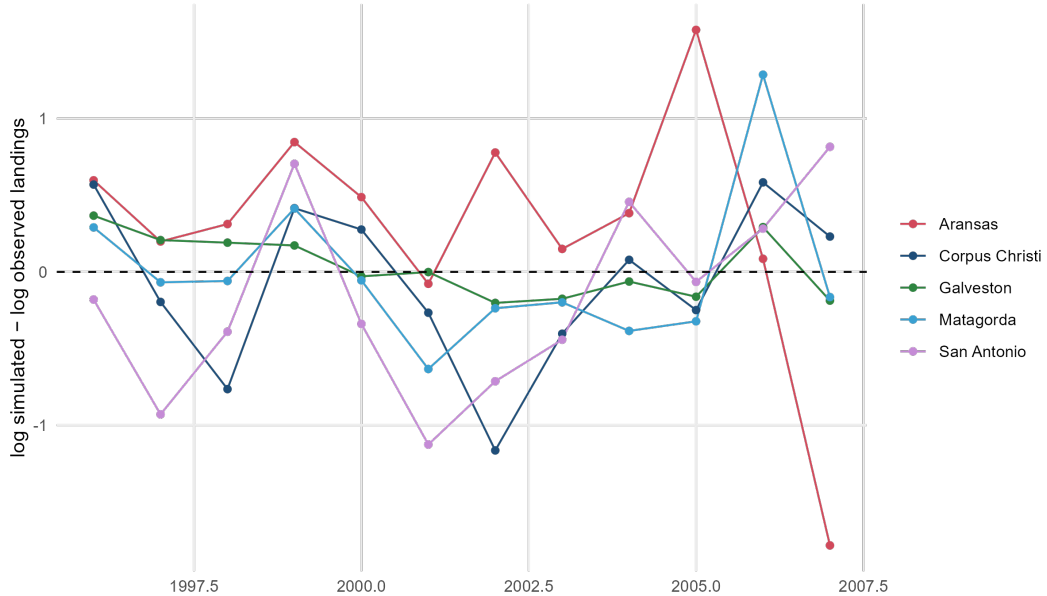


Figure B.1: **Back-cast calibration residuals.** Log simulated minus log observed in-season landings by bay and season, 1996–2007, at the calibrated back-cast (treated arm, deterministic flows).

Table B.1: Calibration of the 1996–2007 back-cast

Parameter	Estimate	SE
Stock trend		
a (1996 level, log)	0.565	0.106
b (annual trend)	-0.0635	0.0153
Bay-level offsets (1996–2007)		
Bay offset: Aransas	0.278	—
Bay offset: Corpus Christi	-0.142	—
Bay offset: Galveston	0.058	—
Bay offset: Matagorda	0.098	—
Bay offset: San Antonio	-0.292	—
Reactivation intercepts		
Reactivation intercept: Aransas	-1.798	0.402
Reactivation intercept: Corpus Christi	0.219	0.167
Reactivation intercept: Galveston	-0.051	0.089
Reactivation intercept: Matagorda	-0.169	0.153
Reactivation intercept: San Antonio	1.392	0.284
Fit: 2008–2023 fleet size		
Active vessels 2008–23, Aransas	16.7 sim / 16.2 obs	
Active vessels 2008–23, Corpus Christi	21.9 sim / 21.7 obs	
Active vessels 2008–23, Galveston	117.9 sim / 118.3 obs	
Active vessels 2008–23, Matagorda	46.6 sim / 46.3 obs	
Active vessels 2008–23, San Antonio	27.7 sim / 28.7 obs	

Note: Joint GMM on the treated arm: log in-season landings by bay-year 1996–2007 (60 moments) and log mean active vessels by bay 2008–2023 (5 moments), diagonal inverse-variance weights. Bay offsets are constrained to sum to zero (the mean is in a). Hansen $J = 23.39$ on 54 df; with heuristic weights and simulated moments J is descriptive. Standard errors from the sandwich with a numerical Jacobian. Search: Nelder–Mead over (a, b) with a fixed-point inner solver for the intercepts.

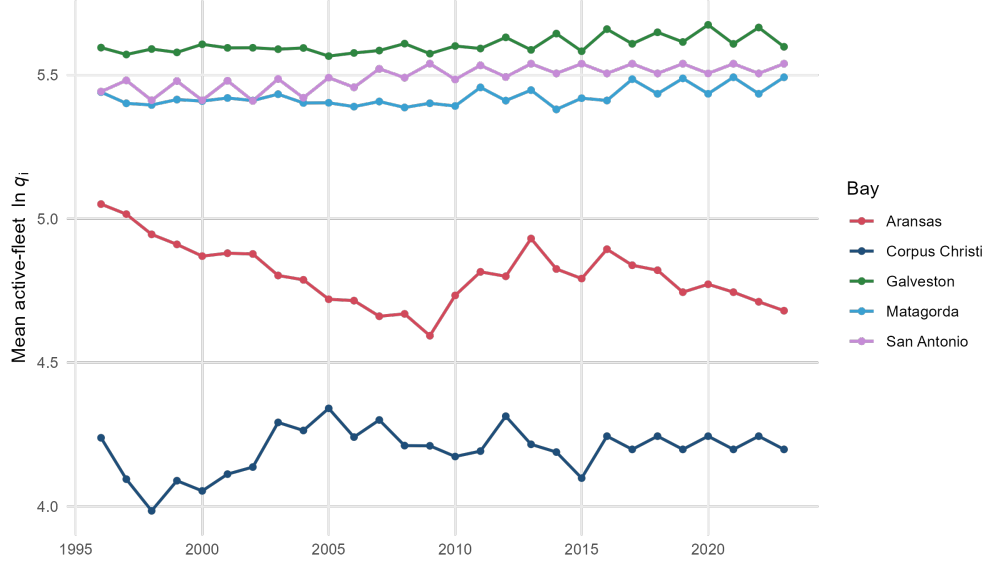


Figure C.1: Mean log-catchability of active vessels, by bay-year

C Inference for the Counterfactual

The participation effect is a function of estimated parameters and of a simulated fleet, so its uncertainty has three sources, reported separately. First, the synthetic fleet is a random draw: re-drawing it 30 times with the calibrated intercepts held fixed moves the estimate between 13.5% and 14.8%, and re-solving the intercepts for each draw (five draws) between 13.2% and 14.7%. Second, the structural parameters and the entry coefficient are estimated: drawing 200 parameter vectors jointly from the structural covariance and the entry coefficient from its clustered standard error, and recomputing the effect for each, gives a 95% interval of 6.0–21.5%; the drawn entry coefficient explains 99% of the variation across draws (correlation -0.99), so the structural parameters contribute almost nothing. Third, because the effect is essentially the entry coefficient’s sampling error pushed through the entry logit, a closed-form band from $\hat{\beta}_{CV} \pm 1.96 \text{ s.e.}$ at the record’s matched ΔCV gives 5.4–21.8%, which is the band used for the sensitivity exhibits. Standard errors of the structural parameters are conditional on the first stage; propagating first-stage uncertainty would widen the second interval, but not the entry coefficient.

D Counterfactual Supplementary Exhibits

Table D.1: Sources of uncertainty in the participation effect

Source	Method	Range
Synthetic fleet draw, intercepts fixed	30 rosters	13.5–14.8%
Synthetic fleet draw, intercepts re-solved	5 rosters	13.2–14.7%
Structural parameters and entry coefficient	200 parametric draws, 95%	6.0–21.5%
Entry coefficient only	closed form, ± 1.96 s.e.	5.4–21.8%

Table D.2: Offsets to a marginal removal, by bay

Bay	Survivors	Own-bay	Cross-bay	Net (harvest)	Net (effort)	Bay-years
Galveston	192	29.8%	-0.4%	29.4%	3.7%	27
Matagorda	80	14.3%	-0.7%	13.6%	1.5%	27
Aransas	56	10.1%	2.6%	12.8%	2.7%	22
San Antonio	33	3.2%	0.4%	3.6%	1.2%	27
Corpus Christi	28	2.5%	-0.5%	2.0%	0.2%	27

Note: Share of a marginal removed vessel’s season harvest (or trawl-days) recaptured by survivors, from drop-one-vessel simulations of the realized-buyback arm, 1997–2023, means over bay-years. Own-bay and cross-bay sum to the net harvest offset.

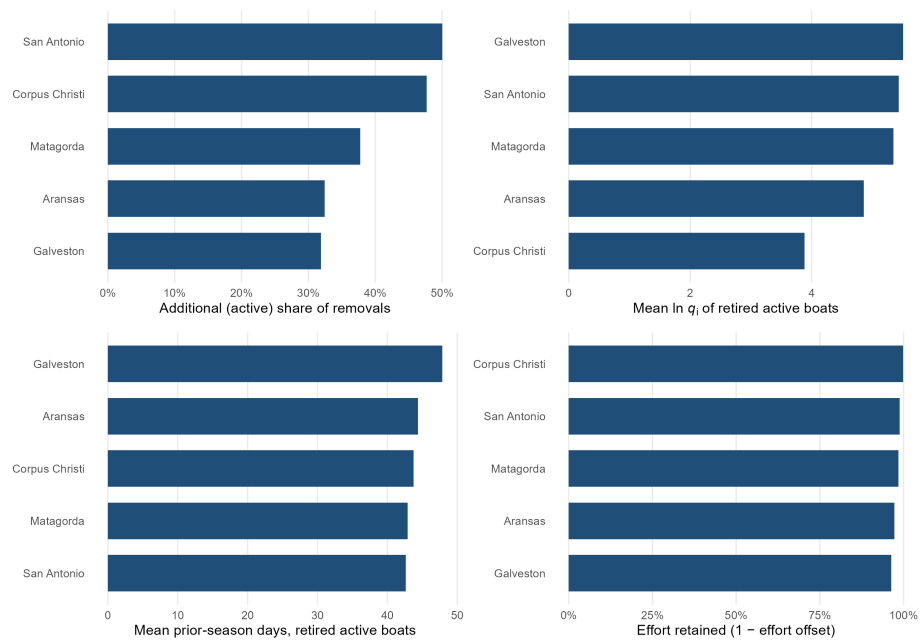


Figure D.1: Offset Decomposition

Notes: Shows the decomposition of what drives per-license conservation by bay. Net trawl-days avoided per license is the product of the additional (active) share of removals, the effort (days fished) of the retired vessels, and net capacity retained (1-offset).

E Model Validation

In this section, I compare the simulated fleet with the observed fleet over 2008-2023 on objects the calibration did not target. The simulation reproduces bay-year landings levels and Galveston’s trajectory ($R^2 = 0.65$ in logs), mean trawl-days per active vessel by bay, and the 2008-2023 fleet sizes it was calibrated to. It misses bay-specific recruitment years, which an expected path simulation smooths. It exits more slowly than the observed fleet after 2017 and reactivates dormant vessels at a third of the observed rate. It removed active vessels fish more prior-season days than the program’s active sellers. Finally, the loyalty mask sets cross-bay fishing to zero in three bays where 3-11% is observed. The last two are bounded in Section 6.6, while the first two bias the participation suppression effect toward zero.

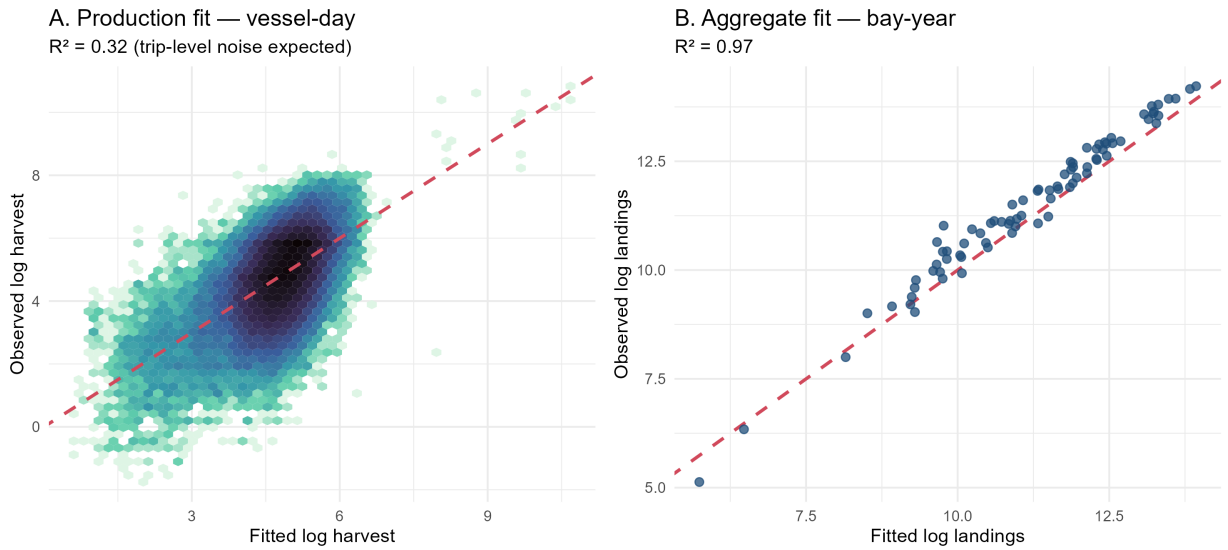


Figure E.1: **Production-function fit at the trip and aggregate levels.** Observed against fitted log harvest; the dashed line is the 45° identity. *Panel A* is at the vessel-day (trip) level, where idiosyncratic trip noise is expected and the fit is correspondingly loose ($R^2 = 0.32$). *Panel B* aggregates to the bay-year level where the stock index and depletion dynamics operate and fits tight ($R^2 = 0.97$).

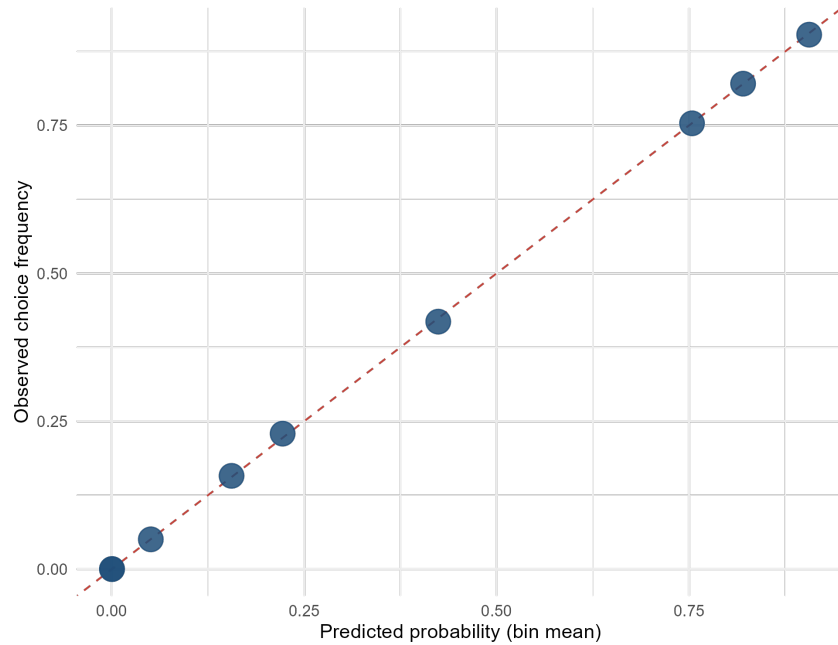


Figure E.2: **Calibration of the first-stage choice probabilities.** Reliability plot for the nested-logit conditional choice probabilities. Vessel-day location choices are grouped into bins by predicted probability; the plot shows the mean predicted probability against the observed choice frequency in each bin (marker area proportional to bin count). Points track the 45° line across the full range, indicating the estimated CCPs are well calibrated, suggesting predicted probabilities match realized choice frequencies at both rare and common alternatives.

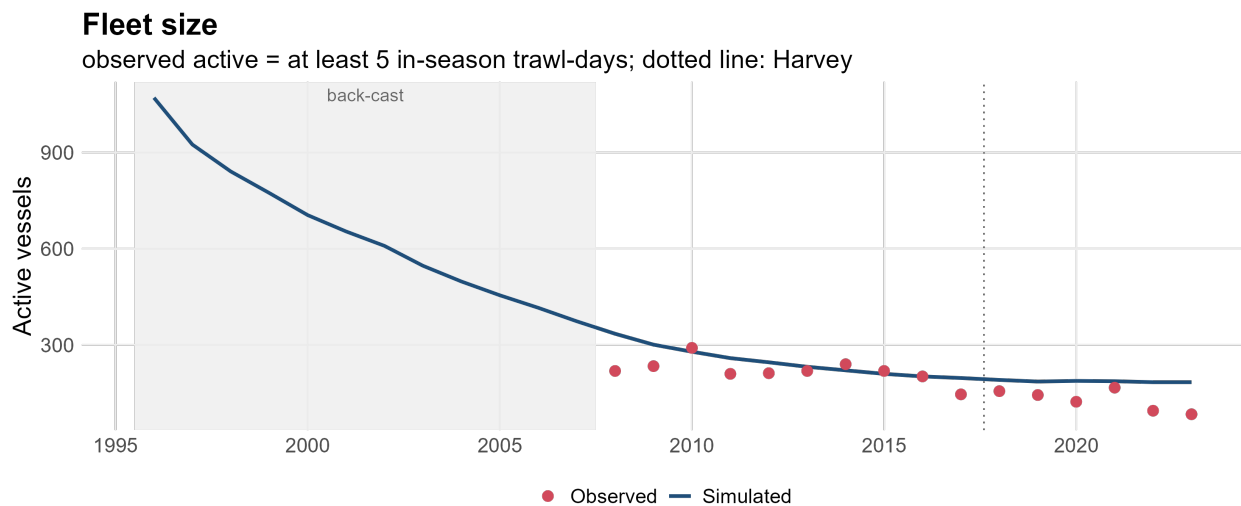


Figure E.3: **Fleet decay: simulated versus observed active fleet.** Overlaid is the simulated active-vessel path (line) on the observed active fleet (points) over the program horizon; the shaded region is the pre-2008 back-cast (calibrated, not estimated, Appendix B) and the dotted line marks Hurricane Harvey (2017).

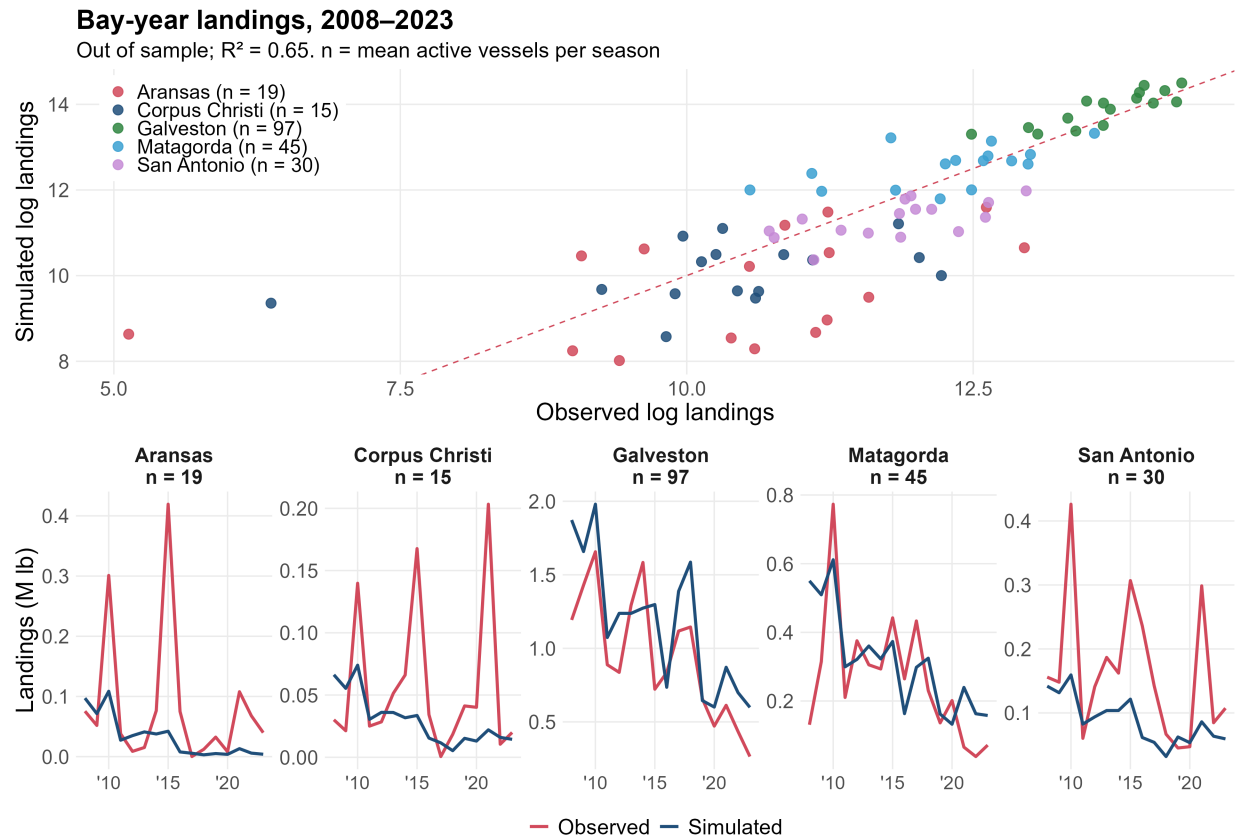


Figure E.4: **Bay-year landings, simulated versus observed, 2008–2023.** Top: simulated against observed log in-season landings by bay-year; dashed line is the 45° identity. Bottom: the two series by year within each bay (million lb).

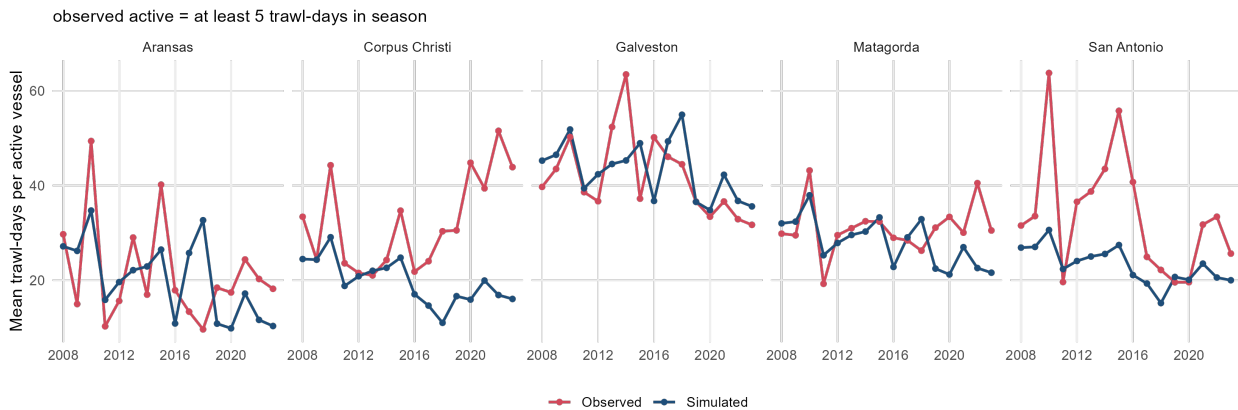


Figure E.5: **Mean trawl-days per active vessel by bay-year, simulated versus observed.** Observed active vessels are those with at least five in-season trawl-days; simulated days are expected days under the deterministic state path.

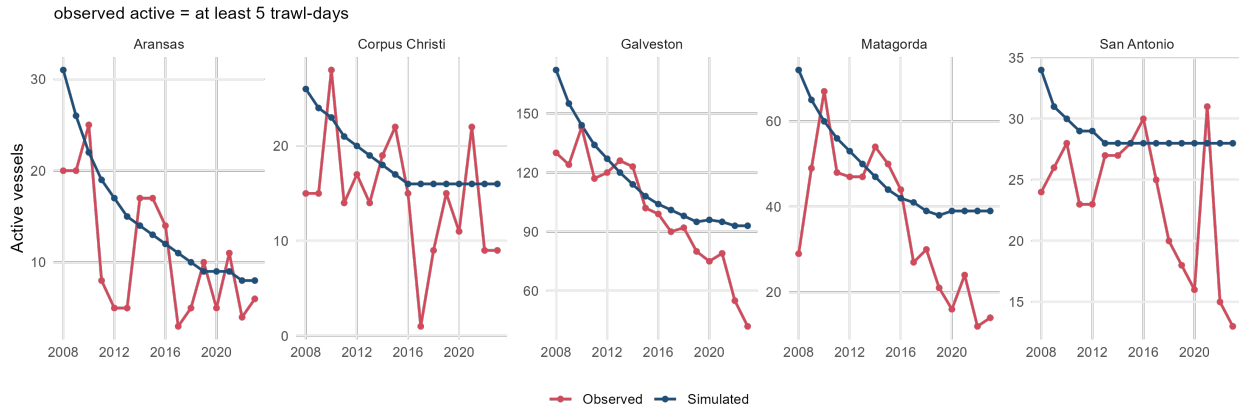


Figure E.6: **Active vessels by bay-year, simulated versus observed.** Observed active vessels are those with at least five in-season trawl-days. The calibration targets the 2008–2023 mean fleet size by bay, not its path.

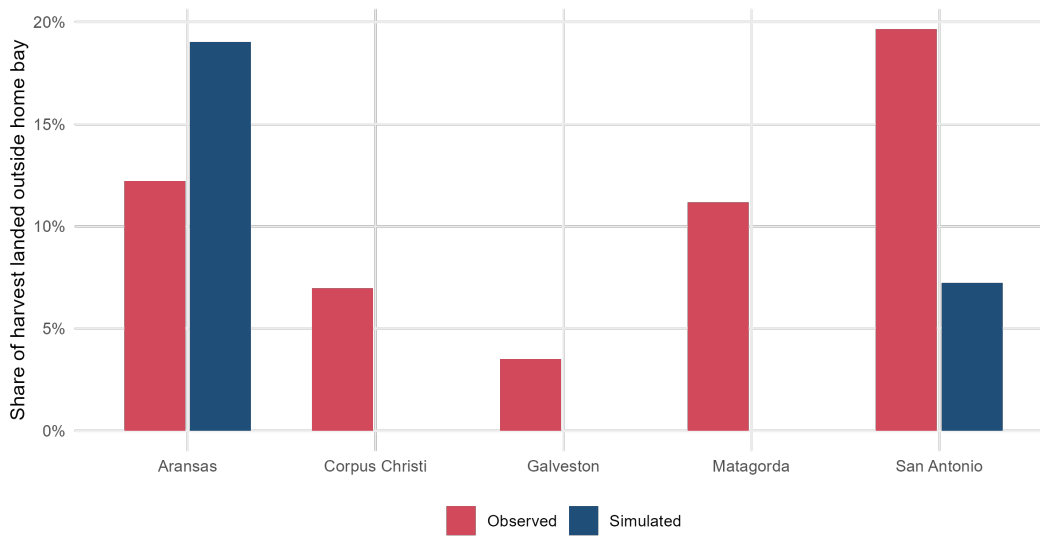


Figure E.7: **Share of harvest landed outside the home bay, simulated versus observed, 2008–2023.** Home bay is the vessel’s modal landing bay. In the simulation, the choice set is restricted by the loyalty mask (Section 6.2).

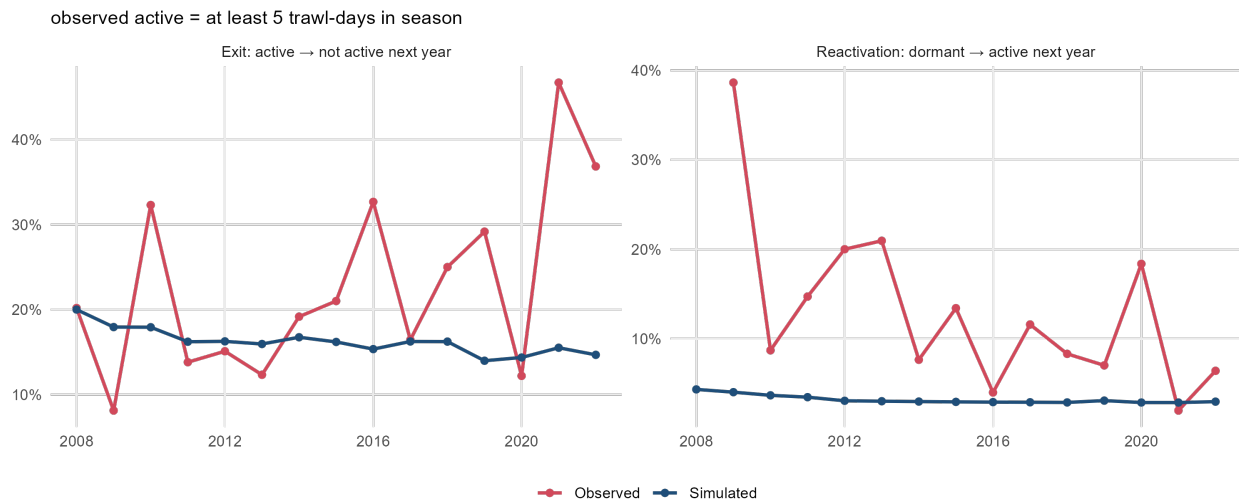


Figure E.8: **Annual exit and reactivation rates, simulated versus observed.** Exit: share of active vessels not active the following year. Reactivation: share of dormant, previously active vessels that are active the following year. Observed activity is at least five in-season trawl-days.

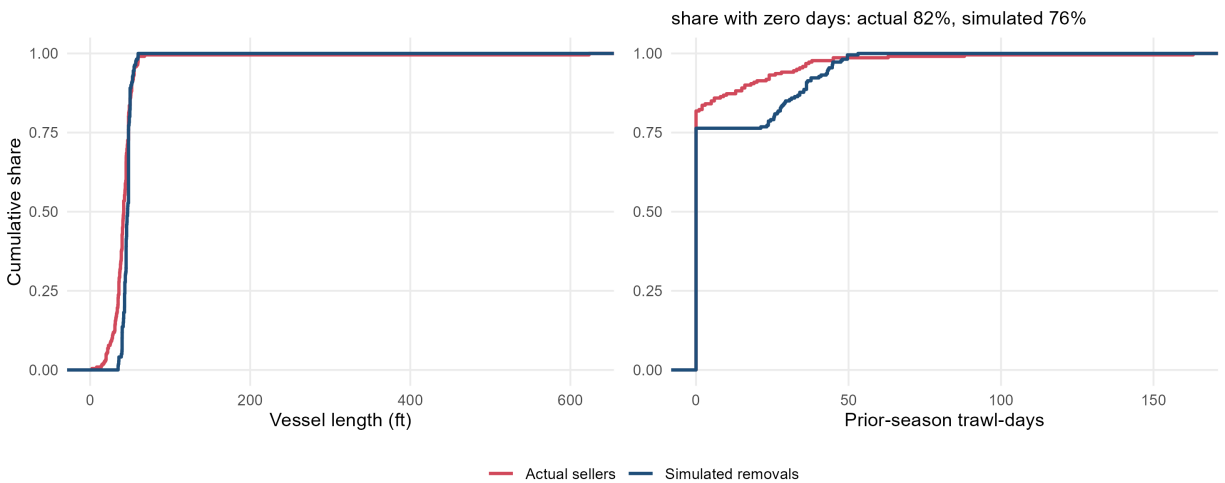


Figure E.9: **Removed vessels: simulated removals versus actual sellers, 2008–2023.** Cumulative distributions of vessel length (left) and prior-season trawl-days (right). Actual sellers are vessels whose bay-license bids TPWD accepted; simulated removals are the active-pool removals in the record arm.

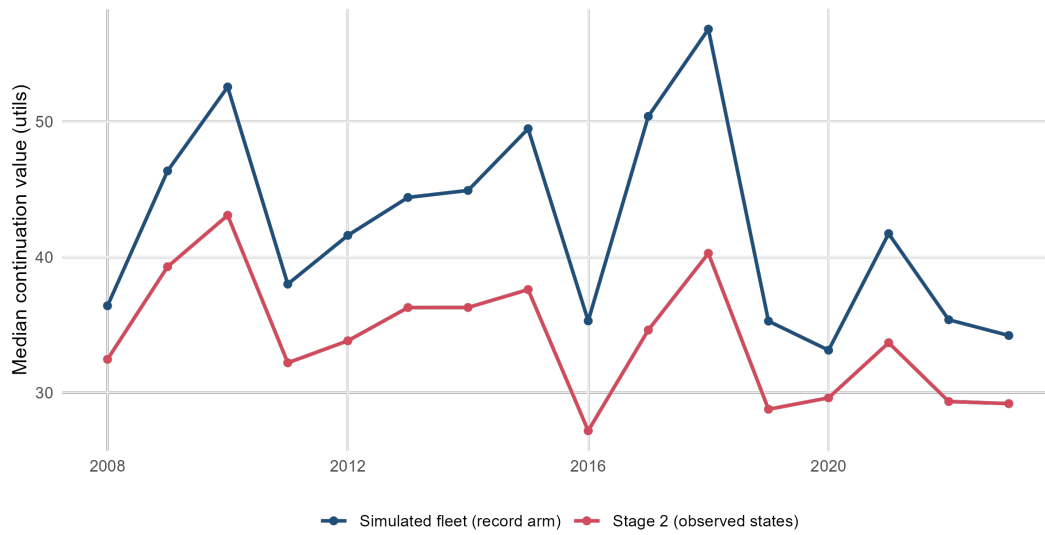


Figure E.10: **Median continuation value by season: synthetic fleet versus Stage 2 estimates.** The synthetic-fleet series is from the record arm; the Stage 2 series is computed on observed vessel states (Section 5.6). Both use the same structural parameters.